81 Oceania Region

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CHAPTER OUTLINE

KEY POINTS

- The burden of chronic kidney disease is increasing across the Oceania region, largely driven by growth in noncommunicable chronic illnesses such as diabetes mellitus.
- Access to health care, ranging from primary preventative management to renal replacement therapy, is limited in many nations of the Oceania region outside of Australia and New Zealand, especially for those who are unable to afford care.
- In response to growth in chronic kidney disease prevalence, countries such as Fiji are currently in the process of expanding their capacity to care for these patients with financial support from their government.

INTRODUCTION

Oceania is a collective name for the land masses scattered throughout most of the Pacific Ocean. It is the term originally conceived by the French explorer Dumon d'Urville (1790–1842) in 1812 as "Océanie," which is a French word derived from the Latin word *oceanus,* meaning ocean. In its broadest sense, the term *Oceania* embraces the entire insular region between Asia and the Americas. The most commonly used definition of the term, however, excludes Ryukyu, Kuril, Aleutian islands, Japan archipelago, Indonesia, Taiwan, and the Philippines, because the people and cultures of those islands are more closely related historically to Asia. Oceania then, in its most commonly defined terms, includes >10,000 islands, with a total area of 8.526 million km^2 and estimated population of >40 million people. Oceania can be divided into four geopolitical subregions: Australasia, Melanesia, Micronesia, and Polynesia (Fig. 81.1), formed by 14 countries and 25 sovereign states and dependent territories, including Australia (population, 23 million), Papua New Guinea (8 million), New Zealand (4.5 million), and other island nations with populations ranging from <100 to >450,000 (Table 81.1).

Indigenous Australians are the original inhabitants of the Australian continent and nearby islands. Although disagreeing on details, scientists generally support a theory that calls for a Southeast Asian origin of the island people in other regions of Oceania. Over the years, many cultures have been influenced by migration, invasion, and colonization with resultant diversity in cultures, economy, language, and religions. Increased influence of the Western world in the region of Oceania has led to an increased urbanization and witnessed a growing burden of adverse health outcomes including obesity, hypertension, diabetes, and chronic kidney disease (CKD).

Kidney disease is one of the major socially transmitted diseases that contributes to higher health-care utilization, poor quality of life, and serious adverse outcomes including cardiovascular disease (CVD), need for renal replacement therapy (RRT), and death.[1,2] Although the exact incidence and prevalence across the Oceania region are incompletely known, reported deaths and years lost due to disability resulting from CKD range up to 4.28% and 2.24%, respectively (Table 81.1). Quantification of the burden of kidney disease is least well understood in the lowest income countries, and given the need for higher health-care utilization, there is an associated cost of kidney disease, which needs to be understood in the context of local government health-care expenditure.

The per capita income [gross domestic product (GDP) nominal per capita], not accounting for the cost of living, ranges widely in Oceania from those countries among the highest incomes in the world (Australia, US$44,820; New Zealand US$38,399) to under US$10,000 in many smaller island countries, states, and territories. The proportion of government expenditure spent on health also ranges from 4% (Papua New Guinea, Vanuatu) to 29% [French Polynesia

Fig. 81.1 Map of Oceania region. (From Geographic Guide. Available from: http://www.geographicguide.com/oceania-map.htm. Last accessed January 24, 2019.)

(France Protectorate)]. Unsurprisingly, there is considerable variation in health outcomes and risk factors for long-term conditions (Table 81.1). This chapter discusses the burden of kidney disease, both acute and chronic, in Australia, New Zealand, and the island nations of Oceania. However, due to the relative paucity of information and data available from the island nations of Oceania, the main focus of this chapter will be on Australia and New Zealand.

AUSTRALIA

BURDEN OF DISEASE

In Australia, >1.7 million adults over the age of 18 years have indicators of CKD, such as reduced estimated glomerular filtration rate (eGFR) or proteinuria.[3] This equates to 1 in 10 Australian adults, with even higher incidence amongst indigenous Australians (1 in 5)[4] who make up approximately 3% of the total Australian population (estimated to be 669,900 people as of June 2011).[5] In contrast to the nonindigenous population where the CKD burden appears to be predominantly situated in major cities, the highest CKD burden in the indigenous population is found in the remote and very remote areas (remote 34% vs. nonremote 13%).[4,6] A small proportion of these patients will progress to end-stage kidney disease (ESKD), requiring RRT. With the growth in CKD burden, the number of patients with ESKD has doubled in Australia over the past decade and is currently >20,000 patients.[7] This growth is not surprising, and is an inevitable consequence of background increase in both behavioral (e.g., inactive/insufficiently active: 56%; smoking on daily basis: 16%) and biomedical [e.g., overweight or obese: 63%; high blood pressure: 32% (including 22% with uncontrolled high blood pressure)] risk factors for CKD in the Australian general population.[6] For instance, the prevalence of diabetes mellitus has tripled between 1989–90 and 2014–15 (1.5% vs. 4.7%), affecting >1.2 million Australian adults.[8] It is the fastest growing chronic disease and diabetic nephropathy has become the most common cause of ESKD in Australia (37%) as a result.[9] Patients with CKD stages 1–5 exhibit risk factors for both CKD and CVD more frequently than people without CKD, including diabetes mellitus (14.2% vs. 6.6%), high blood pressure (39.1% vs. 27.8%), smoking (17.6% vs. 15.7%), and obesity (25.7% vs. 20.0%).[10] Inherently, the burden of CKD relates not only to the health-care requirements of patients who progress to ESKD, but also to broader public health through its close association with CVD, which affects 22% of Australian adults.[11] CKD was estimated to cause 2.6% of the total burden of disease and injury in Australia in 2003,

Table 81.1 Population and Gross Domestic Product (GDP) per Capita of Oceania Countries[176–178]

Country	Population (in thousands)[a]	GDP per Capita (US$ per person/ year)	General Government Expenditure on Health as % of Total Government of Expenditure	Life Expectancy at Birth (Years)	Under-Five Mortality Rate (Probability of dying by age 5 per 1000 live births)	Raised Blood Pressure (%)[b]	Obesity (%)[b]	Deaths Attributable to CKD (%)	Years Lost Due to Disability Attributable to CKD (%)	Disability Adjusted Life-Years Attributable to CKD
American Samoa (USA)	57	9164	14	73	9.4	n/a	n/a	n/a	n/a	n/a
Australia	23,470	44,820	17	73	4.0	21.4	26.8	2.38 (1.99–3.45)	2.05 (1.63–2.57)	1.85 (1.6–2.21)
Cook Islands (NZ)	15	19,659	14	72	10.3	32.4	63.7	n/a	n/a	n/a
Federated States of Micronesia	103	2223	20	70	39.0	n/a	n/a	n/a	n/a	n/a
Fiji	859	6557	9	68	17.9	28.9	30.6	2.97 (2.64–3.92)	1.07 (0.84–1.38)	2.42 (2.09–3.16)
French Polynesia (France)	270	16,803	29	75	2.2	n/a	n/a	n/a	n/a	n/a
Guam (USA)	175	22,991	9	79	0.7	n/a	n/a	n/a	n/a	n/a
Kiribati	109	1701	10	75	47.3	23.6	46.0	2.97 (1.55–6.43)	0.92 (0.71–1.19)	2.24 (1.24–4.53)
Marshall Islands	54	2851	17	72	6.0	22.8	45.4	4.28 (1.74–12.16)	1.01 (0.79–1.31)	3.12 (1.4–8.12)
Nauru	11	5462	10	61	37.0	31.1	71.1	n/a	n/a	n/a
New Caledonia (France)	260	36,637	10	77	5.0	n/a	n/a	n/a	N/a	n/a
New Zealand	4242	38,399	10	n/a	6.6	21.6	28.3	2.43 (1.98–3.19)	2.24 (1.74–2.84)	2.14 (1.86–2.52)
Niue (NZ)	1	21,043	18	74	16.1	n/a	n/a	n/a	n/a	n/a
Northern Mariana Islands (USA)	56	11,719	25	n/a	8.2	n/a	n/a	n/a	n/a	n/a
Palau	18	8423	11	64	12.2	26.6	48.9	n/a	n/a	n/a
Pitcairn Islands (UK)	0.05	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Papua New Guinea	7275	2133	4	62	75	16.7	16.2	1.55 (1.05–2.18)	0.92 (0.7–1.21)	1.2 (0.82–1.68)
Samoa	188	408	7	74	19.4	30.6	54.1	3.66 (1.22–9.81)	1.18 (0.95–1.47)	2.63 (1.18–6.09)
Solomon Islands	611	1954	9	67	15.5	21.2	30.0	2.09 (1.59–2.98)	1 (0.78–1.28)	1.71 (1.28–2.35)
Tokelau (NZ)	1	1007	10	n/a	0.0	n/a	n/a	n/a	n/a	n/a
Tonga	103	7738	5	71	23.0	30.0	57.6	1.69 (0.94–4.03)	1.11 (0.92–1.4)	1.27 (0.85–2.57)
Tuvalu	11	3348	17	70	10.3	n/a	n/a	n/a	n/a	n/a
Vanuatu	265	295,349	4	71	30.7	33.8	27.5	1.39 (0.93–2.12)	1 (0.76–1.29)	1.17 (0.81–1.77)
Wallis and Futuna (France)	12	3800	24	76	n/a	n/a	n/a	n/a	n/a	n/a

[a]Data from World Health Organization, Health Information and Intelligence Platform (HIIP). Available from: http://hiip/wpro/who.int/portal/Countryprofiles.aspx. [Accessed December 2016].
[b]Data from World Health Organization, Noncommunicable diseases and mental health. Available from: http://www.who.int/nmh/countries/en/. [Accessed December 2016].
CKD, Chronic kidney disease; *n/a,* not available.

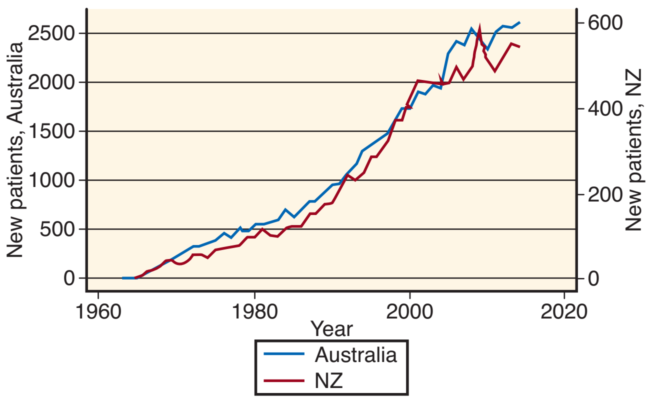

Fig. 81.2 Number of patients starting renal replacement therapy in Australia and New Zealand (NZ). (From ANZDATA Registry. Chapter 1: Incidence of end-stage kidney disease. In: *38th Report*. Adelaide, Australia; 2015. Available from: http://www.anzdata.org.au. Last accessed December 10, 2018.)

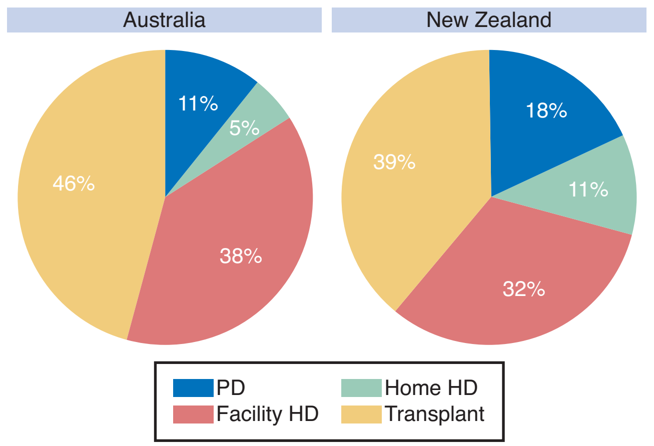

Fig. 81.3 Overall distribution of renal replacement therapy (RRT) modality in Australia and New Zealand at the end of 2015. *HD,* Hemodialysis; *PD,* peritoneal dialysis. (From ANZDATA Registry. Chapter 2: Prevalence of end-stage kidney disease. In: *39th Report.* Adelaide, Australia; 2016. Available from: http://www.anzdata.org.au. Last accessed December 10, 2018.)

with nearly 95% of this burden attributed to years of life lost due to premature death.[10] Patients with CKD incur 85% higher health-care costs and 50% higher government subsidies than individuals without CKD. In 2012, the total costs attributable solely to predialysis CKD were estimated at $4.1 billion, made up of $2.5 billion in direct health-care costs, $700 million in direct non–health-care costs, and $900 million in government subsidies.[12,13] The cost of treating patients with ESKD with dialysis and transplantation are estimated to cost >$1 billion/year.[13]

DIALYSIS

According to the Australia and New Zealand Dialysis and Transplant (ANZDATA) Registry, there were 2654 new RRT patients in Australia in 2015 (Fig. 81.2), with an overall incidence rate of 112 per million population (pmp). Although this rate has remained stable for several years, the number of prevalent patients has continued to climb with 23,012 (968 pmp) patients receiving RRT at the end of 2015.[9] This figure contrasts with the large variation in ESKD prevalence across the world, which is highest in Taiwan (3170 pmp), followed by Japan (2620 pmp), and then the United States (2080 pmp). In the 28 countries that form the European Union, it averages to 1090 pmp with the global average of 450 pmp.[14] A large discrepancy in prevalence of ESKD across regions may relate to variation in population characteristics, but also suggests limited access to treatment in many countries.

The mean ages of female and male patients commencing RRT were 55.7 years and 60.9 years, respectively. The most common cause of ESKD amongst incident RRT patients in Australia is diabetes mellitus (37%), followed by glomerulonephritis (18%), hypertension (13%), polycystic kidney disease (6%), reflux nephropathy (2%), and others.[9] In 2015, over half (54%) of ESKD patients received dialysis (Fig. 81.3), which comprised facility hemodialysis (38%), peritoneal dialysis (PD; 11%), and home hemodialysis (5%).[15]

The first dialysis in Australia took place at the Royal Brisbane Hospital on February 10, 1954, under the supervision of Dr. Dique, who developed a dialysis machine consisting of cellophane tubing wound around wooden slats.[16] The number of patients receiving hemodialysis has grown over the years. In 2015, there were a total of 9947 patients who received hemodialysis as their RRT, with 2195 patients who were new to hemodialysis.[17] In Australia, hemodialysis can be accessed in two main settings: facility (88%) or at home (12%).[15] Facility hemodialysis can be accessed either in hospital (also known as in-center) or in a satellite unit, which is often a regional hospital or community-based dialysis unit distant from the tertiary hospital unit. Increases in the growth of satellite units and PD have contributed to a gradual reduction in the proportion of patients undertaking home hemodialysis in Australia (50% of dialysis patients on home hemodialysis in the mid-1970s, to about 10% from the early 2000s).[18] Although it appears to be relatively small compared with the proportion of patients undergoing facility hemodialysis, the prevalence of home hemodialysis in Australia is second in the world after New Zealand (15.6%).[19] There are regional differences in the uptake of home hemodialysis programs in Australia, whereby the numbers are greatest in the state of New South Wales (including Australian Capital Territory, n = 2700), followed by Queensland (n = 284), Victoria (n = 214), and Western Australia (n = 81). The numbers of prevalent home hemodialysis in other states were <40 patients.[17] The reasons behind such large disparities remain unclear.[18]

There are also opportunities to help facilitate travel on dialysis across Australia, including prearrangement with the dialysis unit near the travel destination, or participating in programs such as "Big Red Kidney Bus" holiday dialysis program, which provides a mobile dialysis service. The bus travels to popular holiday destinations where it is located for up to 6 weeks at a time, and is staffed by experienced dialysis nurses and renal technicians. This is an initiative of the Kidney Health Australia currently available in two states of Australia (Victoria and New South Wales).[20]

Hemodialysis patients in Australia most frequently undergo treatment ≤3 sessions/week (86%), 5 hours/session (26%).[17]

This equates to a mean treatment duration of 271.57 minutes/session, which is the longest amongst countries (compared with US 217.42 minutes/session) contributing data to the Dialysis Outcomes and Practice Patterns Study (DOPPS).[21] Analysis of the DOPPS data has shown lower mortality risk for patients undergoing longer dialysis treatment duration – hazard ratio for every 30 minutes: all-cause mortality: 0.94 [95% confidence interval (CI): 0.92–0.97], cardiovascular mortality: 0.95 (95% CI: 0.91–0.98), sudden death: 0.93 (95% CI: 0.88–0.98).[22] Over the years, the uptake of hemodiafiltration has increased markedly, now taking up almost 21% of hemodialysis treatment[17] (Fig. 81.4). Using these treatment measures and dialysis prescription, 90.9% of patients achieve a urea reduction ratio ≥65% as a measure of dialysis adequacy. This achievement has been relatively stable over the years.[17]

Most prevalent hemodialysis patients (about 85%) in Australia dialyze via either arteriovenous fistula or graft.[17,23] Disappointingly, however, the majority of incident ESKD patients start hemodialysis with a catheter, which is associated with poorer patient outcomes including increased infection, hospitalization, and mortality rates compared with arteriovenous fistula or graft.[24] The risk was greater in patients who were female, young (age <25 years), and first seen by nephrologists <3 months prior to starting hemodialysis (late referrals).[17] Several clinical networks at a state level (e.g., Victoria and South Australia) have set permanent vascular access at dialysis initiation as one of the measured key performance indicators to increase its rates.[25,26]

Patient survival on hemodialysis has remained relatively unchanged over the last few years, with unadjusted survival rates of 91% (95% CI: 90–92), 71% (95% CI: 70–73), and 54% (95% CI: 52–56) at 1, 3, and 5 years, respectively.[17]

The ongoing costs of caring for a patient with ESKD on dialysis are significant, and are greatest for in-center hemodialysis ($76,881/person per year), followed by satellite hemodialysis ($63,505), PD ($51,640), and home hemodialysis ($47,775).[27] With the growth in the prevalent ESKD population, facility hemodialysis capacity has been saturated, and attention has been drawn to enhancing the capacity of home dialysis, particularly PD, which makes up 68% of all home dialysis patients in Australia. The number of prevalent PD patients has grown over time from 93 pmp in 2011 to 106 pmp in 2015.[28] This rate is lower than Hong Kong (488.5 pmp) but greater than many other developed countries (e.g., Germany 38.8 pmp, UK 74.8 pmp, US 87.1 pmp).[29] PD is an attractive dialysis modality due to its technical simplicity, cost-effectiveness, and ability to allow patients to dialyze at home. The ability to independently dialyze at home in a country such as Australia is highly desirable given its large geographic distribution in which patients may live quite distantly from their renal unit. The majority of patients starting PD are incident patients (74%), with smaller proportions transitioning from hemodialysis (24%) or failed kidney transplant (2%).[28] Automated PD (APD) is the most commonly utilized form of PD (70%), and is higher than average APD utilization amongst developed countries (42.4%, 95% CI: 34.4–50.5).[29] Other notable PD practices include a growing usage of icodextrin (46% in 2015 compared with 37% in 2010) and neutral pH, low glucose degradation product PD solutions (18% in 2015 compared with 4% in 2010).[28,30]

Although patient survival on PD is comparable with hemodialysis [1 year 96% (95% CI: 95%–96%), 3 years 79% (95% CI: 75%-81%), 5 years 63% (95% CI: 60%–65%)], further uptake of PD has been hampered by relatively poor technique survival [3 years 49% (95% CI: 45%–53%)].[28] Technique survival has improved slowly over the years, but remains relatively low, and is driven by infection (22%; e.g., peritonitis) and inadequate dialysis (13%; e.g., inadequate fluid ultrafiltration, solute clearance). In recognition of the high burden from peritonitis, the Australian Peritonitis Registry was set up in October 2003 as part of the ANZDATA Registry. This initiative, in conjunction with augmentation of the existing evidence by conducting investigator-initiated clinical trials, ongoing analysis of the data from the ANZDATA registry, improvement in existing guidelines, development of key performance indicators to meet evidence-based practice, and reinforcement of PD training, has witnessed a significant improvement in PD peritonitis rates[31] (Fig. 81.5). However, there remains a large (sevenfold) variation in peritonitis rates amongst PD units and up to twofold variation between states (Fig. 81.6). Although some of these differences relate to patient-level risk factors (e.g., living distantly from PD unit,[32] indigenous[33]), the majority of the risk has been shown to be associated with modifiable center-level characteristics

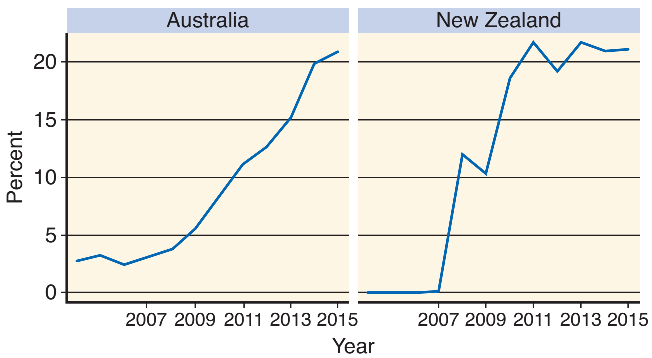

Fig. 81.4 Proportion of maintenance hemodialysis patients receiving hemodiafiltration in Australia and New Zealand 2006–2015. (From ANZDATA Registry. Chapter 1: Incidence of end-stage kidney disease. In: *39th Report*. Adelaide, Australia; 2016. Available from: http://www.anzdata.org.au. Last accessed December 10, 2018.)

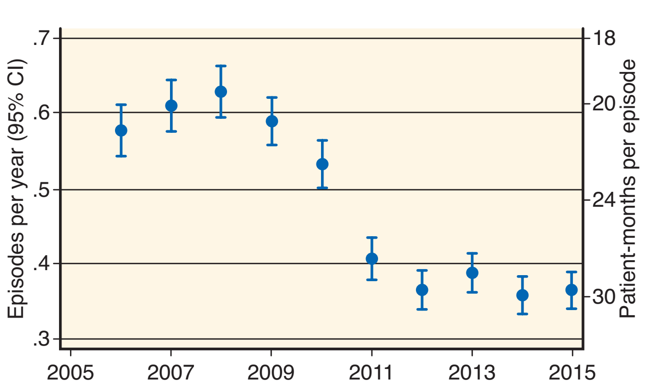

Fig. 81.5 Peritoneal dialysis (PD) peritonitis rates in Australia. *CI,* Confidence interval 2006–2015. (From ANZDATA Registry. Chapter 5: Peritoneal dialysis. In: *39th Report.* Adelaide, Australia; 2016. Available from: http://www.anzdata.org.au. Last accessed December 10, 2018.)

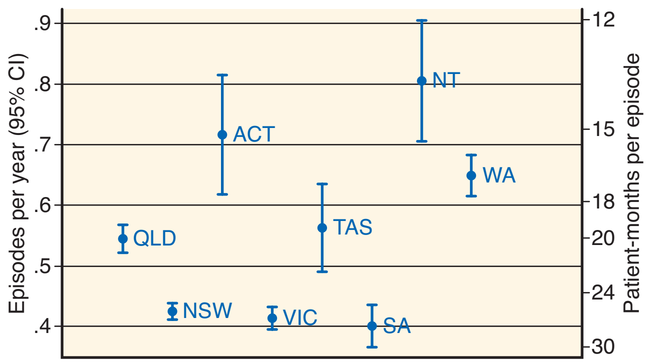

Fig. 81.6 Peritoneal dialysis (PD) peritonitis rates across states and territories in Australia 2006–2015. *ACT,* Australian Capital Territory; *CI,* confidence interval; *NSW,* New South Wales; *NT,* Northern Territory; *QLD,* Queensland; *SA,* South Australia; *TAS,* Tasmania; *VIC,* Victoria; *WA,* Western Australia. (From ANZDATA Registry. Chapter 5: Peritoneal dialysis. In: *39th Report.* Adelaide, Australia; 2016. Available from: http://www.anzdata.org.au. Last accessed December 10, 2018.)

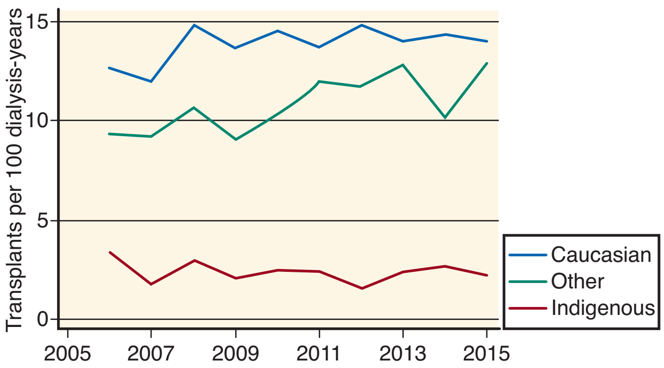

Fig. 81.7 Transplant rate of dialyzed patients in Australia aged 15–64 years by race from 2006 to 2015. (From ANZDATA Registry. Chapter 8: Transplantation. In: *39th Report*. Adelaide, Australia; 2015. Available from: http://www.anzdata.org.au. Last accessed December 10, 2018.)

(e.g., center size, proportion of PD, peritoneal equilibration test use at PD start).[34]

KIDNEY TRANSPLANTATION

In 2015, 949 transplants were performed in Australia, which is the highest number ever performed [40 pmp, which is still only about 30% of the rate of incident ESKD patients (112 pmp)[9]].[35] The transplant rate in Australia is higher than the global average (11.08 pmp) and the highest amongst Western Pacific countries (average 5.82 pmp).[36] Approximately 26% of cases were living-donor transplants ($n = 242$).[35]

The first renal transplant in Australia took place on February 21, 1965, in Adelaide, South Australia. The number of renal transplants has consistently increased since then (2004: 650 patients; 2015: 949 patients), largely from an increase in the rate of deceased-donor renal transplantation in the context of a national reform program "Donate Life" led by the Australian Organ and Tissue Donation and Transplant Authority (AOTDTA) to increase organ donation rate (www.donatelife.gov.au). With the growing burden of CKD and patients with ESKD requiring RRT, there has been a focus on increasing transplant rates as it confers benefits at both patient (e.g., improved survival)[35] and government levels [e.g., cost-effective treatment, estimated cost in the first year $65,375 (deceased-donor renal transplant) to $70,553 (live-donor renal transplant), with ongoing cost from the years thereafter of $10,749/year)].[10] An increase in transplant rate has disproportionately increased in Caucasian patients, while the rate has remained relatively static for indigenous dialysis patients (Fig. 81.7).[35] Similar within-country disparities in transplant rates involving ethnic minorities have been reported previously in Canada[37] and the United States.[38]

Another initiative of the AOTDTA to increase the number of living-donor kidney transplants is the Australian paired kidney exchange program. A paired kidney exchange occurs when a live donor wants to donate to a spouse, friend, or relative but is unable to due to blood or tissue incompatibilities. The program aims to increase living-donor kidney transplants by finding compatible donors among other registered pairs, enabling transplants to occur. The Australian kidney exchange started enrolling donor–recipient pairs in August 2010 and, up until December 2015, 154 people had received a transplant through the program.[39]

An increased rate of kidney transplantation has translated into a growth in the prevalence of patients with functioning grafts to 10,551 patients (444 pmp) in 2015 compared with 6981 patients (341 pmp) in 2005.[35] Similar to dialysis data, there is a variation in prevalence of transplant patients across states ranging from 385 pmp (Western Australia) to 571 pmp (South Australia/Northern Territory). The proportion of patients receiving living-donor grafts was greatest in the younger age group (5–14-year-old group: 66% living donor; 34% deceased donor). For the first time for recipients of kidney transplants in Australia the 1-year graft survival rate was excellent, especially for living-donor kidney transplantation recipients (98% vs. 94% in deceased-donor recipient).[35] The 5-year survival rates for primary transplants performed during 2000 to 2004 were around 87.7% for living-donor transplants and 80.9% for deceased-donor transplants.[27] By 2010–2014, these rates had slightly improved to 89% and 83%, respectively.[35] Although the long-term outcomes (i.e., >10-year graft survival) have also improved gradually, the trend has been less evident compared with short-term outcomes.[35,40] These results similarly mirror data from the United States[41] and the United Kingdom.[42]

According to the ANZDATA Registry, the most common cause of graft loss during the years 2011–2015 was death with a functioning graft ($n = 1049$, 46%), followed by chronic allograft nephropathy ($n = 832$, 37%). The most common cause of death was cancer ($n = 309$, 29%), followed by CVD ($n = 268$, accounting for 26% of deaths).[35] In contrast to a decreasing trend in cardiovascular deaths over the last 20 years from 1.8 deaths/100 patient-years in the mid-1980s to 0.9 deaths/100 patient-years in the mid-2000s, deaths from cancer have progressively increased, which is now the most common cause of death in renal transplant recipients.[43] This result is not surprising as recipients of solid organ transplants have been reported to experience cancer rates similar to nontransplant people 20–30 years older.[44] Reasons for the increased risk of cancer posttransplant have been attributed to several factors, including exposure to new and previously acquired viral infections and ongoing exposure to immunosuppression.[44] Although the patient survival rates are superior after transplant compared with remaining on dialysis, the risk of death remains significantly greater than in the general population by up to tenfold.[40]

NONDIALYSIS SUPPORTIVE CARE FOR ESKD

As aforementioned, the number of ESKD patients is growing in Australia. It is, however, important to note that the growth in the past decade has been the greatest amongst those who are elderly with multiple comorbidities.[45,46] This change has led to an increase in the age of prevalent ESKD patients receiving RRT in Australia, with patients over the age of 65 years making up 39% of the total RRT population.[15] However, the exact number and characteristics of patients who choose not to dialyze when they reach ESKD is unknown, as the ANZDATA registry does not capture information on patients not receiving RRT. The Patient Information about Options for Treatment (PIVOT) study observed that 14% of incident stage V CKD patients chose a nondialysis pathway[47]; however, this is likely to be an underestimate of true disease burden, as it did not account for patients who were never referred to nephrology services in the first instance. Although incompletely understood, it has been suggested that for every patient who dies while receiving RRT, there is another patient who has died without accessing RRT.[10,48]

ACUTE KIDNEY INJURY

Acute kidney injury (AKI) is a condition where an abrupt loss of kidney function occurs to a point where the body accumulates waste products and becomes unable to maintain electrolyte, acid–base, and water balance. Although it has a broad range of causes, it is most commonly diagnosed in general hospitalized patients in developed countries such as Australia (e.g., critically ill patients admitted to intensive care units in the context of sepsis, or following exposure to radiographic contrast agents). Another cause of AKI relatively unique to Australia includes snakebite caused by hemotoxic or myotoxic snakes, such as brown snakes, which account for over half of all hospitalized snakebites in Australia.[49–51] There have been reports of thrombotic microangiopathy-like presentations following brown snakebites with resultant severe AKI requiring temporary treatment with hemodialysis and plasma exchange.[49]

Although CKD is a risk factor for AKI,[52] AKI is also recognized to be an independent risk factor for CKD.[53] In Australia, hospitalizations due to AKI have more than doubled between 2000–2001 and 2012–2013 (from 8050 to 18,010), representing an increase of 6%/year.[54] Even though the deaths due to AKI have remained relatively similar over the past decade (an average of 4670 deaths/year between 2000 and 2012), the average length of stay in AKI hospitalizations was about twice as long as the average length of stay for hospitalizations overall (11.4 days vs. 5.6 days).[54] People living in remote areas, in socioeconomically disadvantaged areas, and with Aboriginal and Torres Strait Islander status were associated with higher rates of hospitalizations and mortality.[54]

INDIGENOUS AUSTRALIANS

Disparity in the risk of renal disease and poorer outcomes are uniformly seen across all phases of disease and treatment in indigenous Australians.[55] For instance, even though indigenous Australians make up only 3% of the total Australian population, almost 11% of new cases of treated ESKD in 2015 were indigenous Australians. The growth in the number of ESKD patients from 2011 to 2015 was almost threefold when compared with nonindigenous Australians (25.7% vs. 8.7%).[55] The hospitalization rate for regular dialysis treatment in indigenous Australians was 11 times as high as for other Australians,[55] which reflects a higher proportion of patients receiving facility hemodialysis (12.7% vs. 51.0%).[56] By contrast, the proportion of prevalent indigenous Australian patients with functioning transplants was only about 2% of all Australian patients (241 patients vs. 10,310 patients).[35] Within indigenous Australians, the risk of CKD-related hospitalizations not involving dialysis was significantly higher in those living in remote and very remote areas (hospitalization rate ratio for indigenous vs. nonindigenous Australians: major city: 2.9; inner regional: 3.8; outer regional: 6.9; remote/very remote: 12.8).[56] Previous studies have shown that remoteness of residence is associated with pervasive socioeconomic disadvantage, including poverty, poor nutrition and food insecurity, poor housing, unemployment, and impaired access to services of all kinds.[57] Furthermore, indigenous Australians were almost four times as likely to die with CKD as a cause of death than nonindigenous Australians.[55]

AKI is also significantly more common in indigenous Australians, with hospitalization rates almost three times higher than those of nonindigenous Australians (males 158 vs. 71 per 100,000 population, respectively; females 229 vs. 70 per 100,000 population).[54] Similarly, AKI death rates are 1.8 times higher in indigenous Australians (35 vs. 19 per 100,000 population) and are considerably higher in indigenous females than indigenous males (2.1-fold vs. 1.6-fold higher than nonindigenous males and females, respectively).[54] As with CKD, there is a complex interaction between indigenous status, remoteness, and socioeconomic status that impacts upon AKI occurrence and outcome in Australia.

ETHNIC INEQUITIES

The burden of kidney disease (AKI, CKD, and ESKD) in indigenous Australians is disproportionately greater compared with that in nonindigenous Australians[55] and affects these patients at an earlier age. For instance, of all the patients who commenced RRT for treatment of ESKD in Australia between the years 2010 and 2014, 59% of indigenous patients were aged less than 55 years, compared with 31% of nonindigenous patients.[58] Although the most common cause of ESKD in Australia is diabetic nephropathy for both groups, the proportion of indigenous patients with diabetic nephropathy is more than double that of nonindigenous patients (70% vs. 33%).[59] Differences in the causes of CKD and AKI across these two groups are more challenging to describe in detail as data are not as readily available due to a lack of established CKD and AKI registries (the ANZDATA only includes ESKD patients[54,58] who received dialysis or kidney transplantation). The reasons for ethnic inequality in the burden of kidney disease amongst indigenous Australians have been attributed to higher levels of socioeconomic disadvantage,[60,61] residence in "remote" and "very remote" areas of Australia,[54] and a high burden of chronic diseases, such as diabetes mellitus which is an important cause of kidney disease.[58] There are also many other contributory factors, including poorer nutrition, higher tobacco use and alcohol consumption, and lower levels of education attainment, employment, social support, and housing.[58] Moreover, there is a greater proportion of indigenous patients who reside in remote and very remote areas of Australia where access to and use of health services are more challenging.[58] In recognition of a significant deficit in health outcomes of indigenous patients in Australia, on December 20, 2007, the Council of Australian Governments agreed to work with indigenous communities to achieve the target of "closing the gap" on indigenous disadvantage. The Closing the Gap campaign aims to address not only gaps in the areas of health, but also in education and employment. One of the three main health goals includes targeting main components of the life expectancy gap and chronic diseases, such as CKD and diabetes mellitus. Some of the listed targets to improve outcomes in this area include attainment of blood pressure targets in patients with type 2 diabetes mellitus, and stabilization of all-cause incidence of ESKD within 5–10 years.[62] In order to help support access to appropriate medications, indigenous patients with chronic diseases or at risk of developing chronic diseases, such as CKD, are eligible to access the Closing the Gap Pharmaceutical Benefits Scheme co-payment measure to help access most prescription medications at a lower price or free of charge, depending on eligibility criteria (http://www.humanservices.gov.au/). Following implementation of the scheme, improvements in health-care access and a declining trend in indigenous mortality rate from chronic diseases have been observed.[63]

DOMAINS OF KIDNEY DISEASE CARE IN AUSTRALIA

HEALTH FINANCE

Australia's health-care system is complex with a multifaceted network of public and private providers. Medical providers deliver a plethora of services across many levels, from public health and preventive services in the community, to primary health care, emergency health services, hospital-based treatment, and palliative care. For most Australians, their first contact with the health system is a visit to a general practitioner (GP) who may identify risk factors or features of renal impairment through clinical assessment. The GP may then refer them to a specialist based in private practice or a public hospital based on clinical findings.

Public sector health services are funded by the state, territory, and Australian governments, but managed by state and territory governments. By contrast, private hospitals are owned and operated by the private sector. Patients who are cared for in public hospitals (e.g., for AKI and CKD) are able to access all investigations and treatments for free if they are eligible for Medicare. Medicare is a scheme administered by the federal government that provides a range of medical services; lower cost prescriptions and free care as a public patient in a public hospital. All eligible Australian residents and certain categories of visitors to Australia can enroll in Medicare and access these services. The Department of Health is responsible for developing Medicare policy, and the fund is partly contributed to by Australian taxpayers, who pay a Medicare levy of 2% of their taxable income (with exceptions for low-income earners), with the balance being provided by government from general revenue. An additional levy of 1% is imposed on high-income earners without private health insurance policies. If patients are treated as a private patient in public or private hospitals, Medicare will cover 75% of the Medicare schedule fee for services and procedures, such as dialysis. However, Medicare will not cover hospital accommodation, or fees such as theater costs. If patients are insured privately through a private health insurance provider, part or all of these costs may be subsidized through the insurance policy in place, including RRT.

ESSENTIAL MEDICATIONS AND TECHNOLOGY ACCESS

As well as Medicare, Australia has a separate Pharmaceutical Benefits Scheme (PBS; www.pbs.gov.au), also funded by the federal government, which subsidizes a range of prescription medications. These two schemes jointly facilitate delivery of universal health-care system in Australia. Eligible indigenous patients who are living with, or at risk of, chronic disease, such as CKD, are able to additionally benefit from the Closing the Gap PBS Co-payment measure which improves access to medicines by further lowering or forgoing patient co-payment for PBS-subsidized medicines. Nonetheless, being a large country with the vast majority of large cities located along the coast where major health facilities reside, there is a significant proportion of the population living in regional areas with limited access to medical services, such as general practitioners (GPs), nephrologists, and renal replacement therapies (RRT), including dialysis and transplantation. According to the ANZDATA Registry, 5.9% of incident PD patients (n = 749) were living in remote and very remote regions, as defined by Accessibility and Remoteness Index of Australia (E. See, personal communication). In recognition of the great need for medical workforce in rural and regional Australia, the Australian government has introduced several initiatives to date, including the Australian General Practice Training Program with rural training pathway, Bonded Medical Places Scheme (i.e., a percentage of all first-year Commonwealth Supported Places in Australian medical schools are allocated to the scheme where successful applicants commit to working in a district of workforce shortage during their return of service period) and 5-Year Overseas Trained Doctor Scheme (which allows a reduction in the 10-year moratorium for overseas-trained doctors by encouraging them to work in remote or difficult-to-recruit locations).[64] These schemes, however, do not address limited access to specialist care or access to RRT closer to home. Patients living remotely,[33] and further from their dialysis providers,[32] have been shown to experience worse outcomes than others with easier access to specialist care. Moreover, if patients require facility hemodialysis treatment, they need to relocate closer to their dialysis unit if the distance is not feasible to travel thrice weekly, which results in significant social dislocation and disruption (see the *Dialysis Unit Guide* by Kidney Health Australia, www.kidney.org.au/your-kidneys/support/dialysis/dialysis-unit-guide).

DETECTION AND MONITORING OF KIDNEY DISEASE

One of the main predictors of poor patient outcomes in those with ESKD has included late referral, defined as less than 3 months between referral and RRT commencement.[9] Although there are cases where prevention or early recognition may not be possible, the majority of CKD cases should have been able to be recognized in primary care and reviewed by a nephrologist prior to imminently requiring RRT. In appreciation of the importance of early recognition of CKD, several initiatives have been developed and evolved over the years in Australia. These have focused on empowering health-care providers to recognize risk factors for CKD to trigger screening, optimally manage risk factors to retard progression of disease, and know when a specialist referral is indicated.

An Australasian Creatinine Consensus Working Group developed recommendations to commence automatic reporting of an eGFR from each serum creatinine measurement performed in pathology laboratories.[65] This was to increase the likelihood of early detection of CKD and indeed witnessed a significant growth in the number of patients diagnosed with CKD since its implementation.[66] There have also been studies to help guide how and who to screen for CKD, such as a pilot program Kidney Evaluation for You.[67] The aim of this study was to establish community-based screening protocols by targeting people identified to be at an increased risk of CKD (i.e., those with diabetes mellitus, hypertension, a first-degree relative with kidney disease, or age >50 years), and to assess efficacy in promoting changes in risk-factor management and to understand the CKD awareness amongst participants. Of 402 high-risk individuals recruited, 20.4% of participants exhibited features of CKD with a large proportion having risk factors for CKD including hypertension (69%), diabetes (30%), and dyslipidemia (40%). Outcomes from such projects have led to recommendations such as the "Kidney Health Check" through Kidney Health Australia, whereby patients with diabetes, hypertension, or other established CKD risk factors are recommended to undergo three tests every year (i.e., blood test to check eGFR, urine test to evaluate for proteinuria or hematuria, and blood pressure measurement).[68]

There has also been a huge focus on augmenting resources and support for GPs who are often the first health-care providers to manage patients with CKD in the community. For example, the Primary Care Education Advisory Committee for Kidney Health Australia formed by nephrologists, GPs, primary health nurses, educators, and government representatives provides free accredited education to GPs to increase their awareness and implementation of best practice detection and management of CKD (www.kcat.org.au). These education sessions are delivered through various platforms, including interactive face-to-face workshops, active learning modules, online learning, and conference sessions.[69] Other strategies to improve quality of care delivery in primary care settings include development of customized software programs to integrate with primary care electronic health-care records to allow real-time promotion of detection and management of CKD according to best practice recommendations (eMAP:CKD),[70] provision of decision support tools for health-care providers (Health Tracker, George Institute),[71] and development of quality improvement programs through the Australian Primary Care Collaborative (http://apcc.org.au).

Although clear guidelines exist on indications for referral to a nephrologist in Australia (e.g., eGFR <30 mL/min/1.73 m^2),[72] some remote living patients are unable to be reviewed easily by nephrologists. In areas where telehealth facilities are available, nephrologists from the nearest renal units have managed selected patients with CKD via telehealth with local nursing support and coordination.[73] These practices have received additional funding support from the Commonwealth Department of Health by reimbursements for telehealth services under the Medical Benefit Scheme. Supporting patients and health-care providers through telehealth has been extended to cover patients receiving RRT in Australia, including those undergoing home dialysis. For example, several renal units have developed phone apps to provide ongoing support, communication, and monitoring for home hemodialysis patients.[74] Renal transplant units may also elect to provide regular telehealth transplant clinic support to regional centers to help improve management of patients and provide support to their clinicians.

HEALTH WORKFORCE

In 2005, the Australian and New Zealand Society of Nephrology (ANZSN) reported 171 full-time–equivalent adult nephrologists (136 full time and 82 part-time) in clinical practice.[75] This equated to 8.6 pmp (or 1 nephrologist for every 116,000 residents) or 1 nephrologist for every 79.5 patients with ESKD.[75] Subsequently, the Australian Nephrology Workforce Survey completed in 2007 identified several factors that could influence the size of the nephrology workforce in future years, including a desire of young nephrologists to retire early, a large proportion of nephrologists who are greater than 55 years of age, and workload and clinical demands that were perceived to exceed personal capacity.[76] Based on these findings, several recommendations were made, including the need to improve recruitment strategies, and to have greater emphasis on regular workforce and workload predictions. Since the release of the report, there has been more than a twofold increase in the number of nephrology training positions in Australia (106 positions in 2014 compared with 56 in 2008). In fact, the projection has overshot the number of nephrologists required to an extent that by year 2018, there will be 94 excess nephrologists (Dialysis Nephrology & Transplant Workshop, Launceston, Tasmania, March 2015).

HEALTH INFORMATION SYSTEMS AND STATISTICS

Information on patients with ESKD receiving RRT is well described through the ANZDATA registry, which collects a wide range of statistics relating to the outcomes of treatment of those with ESKD from all renal units in Australia and New Zealand (www.anzdata.org.au). Its main results are published annually in the form of a report, with interim analyses and publications that occur throughout the year as conference papers or research projects designed to answer specific questions. The data relating to CKD are also collected through the Australian Institute of Health and Welfare (AIHW) through the National Centre for Monitoring Vascular Diseases (www.aihw.gov.au), and Australian Health Survey from the Australian Bureau of Statistics (http://www.abs.gov.au/australianhealthsurvey). The AIHW has also started providing information on AKI burden in Australia, and their official first report was published in 2015.[54] In order to further enhance understanding in disease and outcomes, several newer registries have been developed over the past decade, including statewide collaborative multidisciplinary research and practice programs in various states across Australia, such as CKD.QLD (Chronic Kidney Disease in Queensland, www.ckdqld.org), the Registry of Kidney Diseases (Victoria; http://rokd.org.au), and for specific disease groups, such as for glomerulonephritis (QLD. GN Registry). One of the main criticisms of these existing health information systems has been an unduly long lag period between an event and its being reported. As the health system increases uptake of technology, including adoption of digital hospitals, the data are increasingly likely to be captured and reported in real-time to more efficiently assist planning.

STRATEGIES AND POLICY FRAMEWORKS

Data collected from health information systems help formulate guidelines and direct strategies and policies to improve patient care. In Australia, one of the most well-recognized national guidelines is the Kidney Health Australia – Caring for Australasians with Renal Impairment (KHA-CARI; www.cari.org.au), which lists a broad range of recommendations on caring for predialysis, dialysis, and transplant patients. To ensure the appropriate usage of KHA-CARI guidelines in clinical practice, the group has also launched several implementation projects, in areas such as iron therapy, hemodialysis vascular access, and prevention of infection in PD patients. Kidney Health Australia has been an active advocate to initiate other national policies including the national renal action plan to reduce Australia's kidney disease burden,[77] and is one of the four members of the National Vascular Disease Prevention Alliance committed to reduce the burden of CVD in Australia. The Australian Government also recognizes the growing burden of CKD and has listed it as one of the chronic conditions in the National Strategic Framework for Chronic Conditions, which aims to reduce the impact of chronic conditions in Australia.[78] At a national level, various working groups have published seminal position statement papers to direct clinical practice, such as the Australasian Creatinine and eGFR Consensus,[65] Australasian Proteinuria Consensus,[79] and Home Dialysis Favored Position.[80] Within each state of Australia, there also exist separate statewide renal networks to inform more relevant practice at a local level.

CAPACITY FOR RESEARCH AND DEVELOPMENT

Australia is a major contributor to research and development amongst countries in Oceania. In order to further strengthen its capacity, a collaborative research program, the Better Evidence and Translation – Chronic Kidney Disease (www.beatckd.org), funded by a National Health and Medical Research Council Program Grant was initiated, which aims to improve the lives of CKD patients by generating high-quality research evidence to inform health-care decisions. The program supports four existing national research and translation platforms, including the ANZDATA Registry, KHA-CARI guidelines, Cochrane Kidney and Transplant Group, and the Australasian Kidney Trials Network (AKTN). Cochrane Kidney and Transplant was initially established in 1997 and the editorial base moved to Sydney, Australia, in 2000. It forms one of 53 Cochrane Review Groups that constitute the Cochrane Collaboration, which is an independent organization dedicated to providing up-to-date, accurate information about the effects of health care. The AKTN is also a not-for-profit collaborative research group that designs, conducts, and supports investigator-initiated clinical trials with the aim of improving life for people living with CKD (www.aktn.org.au).[81] Since its inception in 2004 and endorsement by ANZSN and KHA, the AKTN has successfully completed eight trials with ongoing growth in productivity. More recently, there has been a growing focus on priorities from all involved in caring for patients with CKD, including patients, caregivers, clinicians, researchers, policy makers, and other stakeholders. This has led to the Standardised Outcomes in Nephrology (SONG) initiative, which aims to establish a set of core outcomes and outcome measures across the spectrum of kidney disease for trials and other forms of research (www.songinitiative.org). Currently, core outcomes for hemodialysis (SONG-HD),[82–84] transplantation (SONG-Tx), PD (SONG-PD), children and adolescents (SONG-Kids),[85] and polycystic kidney disease (SONG-PKD) are being developed.

NEW ZEALAND

Aotearoa/New Zealand is situated 2000 km southeast of Australia. Comprising two large islands and several smaller islands, New Zealand was first settled following migration across the Pacific Ocean in the late 13th century by Māori, the indigenous population. Māori settled throughout Aotearoa, living within iwi (tribes), hapū (subtribes), and whānau (extended families). Abel Tasman was the first European explorer known to have reached New Zealand in 1642. Progressive settlement by European settlers occurred in the 18th century for whaling and sealing. Progressive colonization in the 19th century by the British Crown resulted in the signing of the Treaty of Waitangi in 1840 between the Crown and iwi. Following World War II, extensive migration to New Zealand by people in the Pacific occurred. Currently, of the 4.6 million people living in New Zealand, 3.3 million (71%) identify as New Zealand European, 692,000 (15%) as Māori, 344,000 (7%) as Pacific people, and 541,000 (12%) as Asian.[86]

ACCESS TO HEALTH CARE

The New Zealand health-care system is predominantly publicly funded by taxes. Twenty District Health Boards (DHBs) are responsible for funding health-care services to their catchment population within a geographical region, providing acute care services, and promoting the health of the population. All public hospital and specialist outpatient services within the public system are free, while attendance in primary care incurs some payment, based on income. Prescriptions provided by hospitals for inpatients and outpatient care in a hospital setting are fully subsidized by the government. RRT (dialysis or kidney transplantation) is provided nearly entirely within the public health-care system. Renal specialists provide dialysis and transplantation care through hospital-based remuneration and are not incentivized to provide specific dialysis treatment modalities. Citizens or permanent residents of New Zealand and dependencies (Niue, the Cook Islands, and Tokelau) receive treatment for ESKD within the tax-funded public health system in New Zealand.

BURDEN OF CHRONIC KIDNEY DISEASE

In Aotearoa/New Zealand, the proportion of the population who have CKD is unknown, as census or systematic observational data have not been reported. Based on 2011–2012 Australian data showing albuminuria present in 7.7% of people aged 18 years or older and 3.6% with an eGFR < 60 mL/min/1.73 m^2,[87] it can be estimated that 361,300 New Zealand adults may have indicators of CKD. The frequency of albuminuria is approximately twofold to fivefold lower for New Zealand European than for Māori and Pacific adults with[88] and without[89] diabetes.

Approximately 500 New Zealand patients (including 20 children and adolescents) commence dialysis for treatment of ESKD each year and 150 receive a kidney transplant, while 472 patients are actively on the waiting list for kidney transplantation.[90] The incidence of RRT in New Zealand increased threefold from 30 pmp in 1986 to 120 pmp in 2001; the incidence has largely plateaued in the last 15 years, with currently 4500 patients receiving treatment. There is considerable regional variation in the incidence of treated ESKD across the country with higher rates in Northern regions, excluding higher-income areas of Auckland.

Diabetes is the most frequent cause of treated ESKD in Aotearoa/New Zealand, with 54% of patients treated with dialysis and 22% of kidney transplant recipients having a diagnosis of diabetes. This proportion is increasing by about 1% each year in the dialysis population and 0.5% each year in the kidney transplant setting. Risk factors for CKD frequently coexist including socioeconomic deprivation, obesity, dyslipidemia, and hypertension.

The burden of disease in New Zealand has become decoupled from disease frequency, reflecting the predominant impact of multiple long-term conditions (multimorbidity) on health burden rather than the presence of individual conditions such as diabetes. About one-third of health lost by New Zealanders is caused by known risk factors for long-term conditions including CKD; high blood pressure accounts for 8.3% of disability-adjusted life years, while low physical activity accounts for 3% of health loss, dietary risks for 9.4%, high body mass index (BMI) for 9.2%, and glucose intolerance for 5.7%. CKD accounts for approximately 2.1% of health loss in New Zealand, ranking 16 in the top 20 leading major specific conditions contributing to health loss for both men and women. CKD is a key outcome from the burden of high BMI, mediated in part through type 2 diabetes.[91]

The annual health expenditure in Aotearoa/New Zealand has risen on average 5.1% compared with annual population growth of 1.3% between 1990 and 2010. The total health expenditure was NZ$19.9 billion in 2009/2010, and in 2016 accounted for 9.2% of real GDP, similar to many peer nations in the Organization for Economic Co-operation and Development (OECD).[92,93] The estimated health costs attributable to CKD in New Zealand have not been specifically reported. Previous studies have estimated the direct costs of diabetes in New Zealand to account for NZ$0.6 billion with an additional indirect cost estimate of NZ$0.6 billion, accounting for 6% of the annual health expenditure.[94] The provision of dialysis is estimated to cost 1% of New Zealand's health expenditure.[95]

INEQUITY BASED ON ETHNICITY

The incidence of treated ESKD in New Zealand is not equitable, with an incidence of 62 pmp for New Zealand European patients, compared with incidences of 216 pmp for Māori and 272 pmp for Pacific patients, giving a 3.5-fold difference in the incidence of treated ESKD. There is considerable regional variation in the incidence of treated ESKD across the country with higher rates in Northern regions, excluding higher-income areas of Auckland. Notably, the age of onset for diabetes diagnosis is 10 years younger among non-European ethnicities in New Zealand, in parallel with the premature onset of treated ESKD at younger age groups for Māori and Pacific patients.

The incidence of dialysis in New Zealand varies over fourfold by ethnicity, with an incidence of 62 pmp for New Zealand European patients, 76 pmp for those identifying as Asian, 216 pmp for Māori, and 272 for Pacific patients. The most common cause of treated ESKD for incident patients in New Zealand is diabetes (39%), followed by glomerulonephritis (17%), hypertension (11%), polycystic kidney disease (5%), and reflux nephropathy (2%).

There is marked inequity in kidney transplantation rates in New Zealand based on ethnicity. Despite an age-standardized risk of ESKD for Māori and Pacific patients, approximately 2 to 10 times that for New Zealand European patients, the rate of kidney transplantation is markedly lower. In 2015, 28 Māori patients received a kidney transplant (38 pmp) while 21 Pacific patients received a kidney transplant (54 pmp), compared with dialysis incidences of 216 pmp and 272 pmp for those populations, respectively. In addition, recent increases in preemptive kidney transplantation appear to be favoring New Zealand European patients. This phenomenon of increased disparity in transplantation rates during periods of improved access through better service delivery has been similarly observed in Australia, Canada, and the United States (Fig. 81.8). In addition, there is considerable variation between

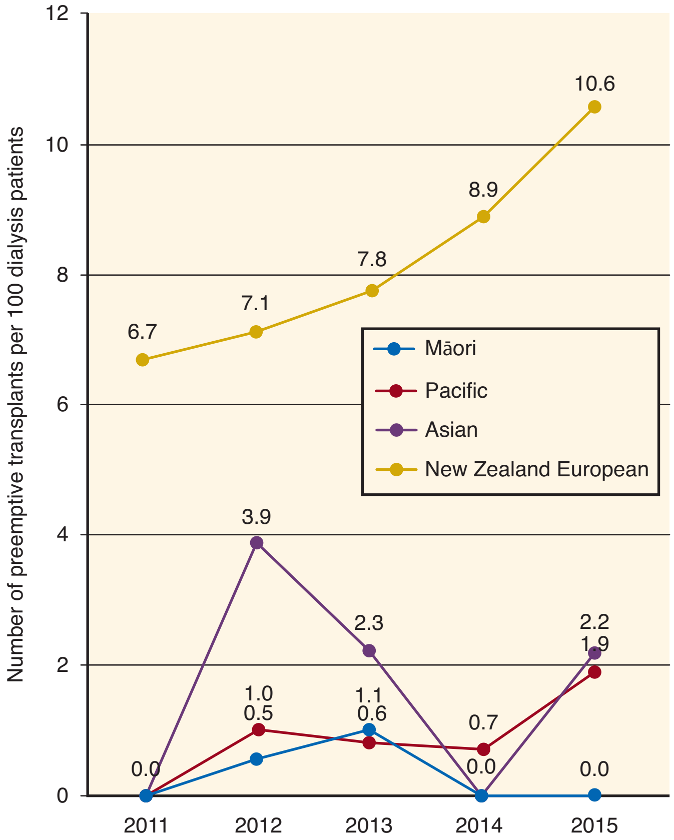

Fig. 81.8 Preemptive kidney transplantation rates in New Zealand by ethnicity, 2011–2015. (From Ministry of Health National Renal Advisory Board. *New Zealand Nephrology Activity Report 2015*. Wellington, New Zealand: Ministry of Health; 2015. Available from: http://www.health.govt.nz/about-ministry/leadership-ministry/expert-groups/national-renal-advisory-board/papers-and-reports. Last accessed December 10, 2018.)

New Zealand renal centers in the rate of transplantation, varying between 9 pmp and 46 pmp in 2015.

DIALYSIS

The population with treated ESKD in Australia and New Zealand has been documented and described within the ANZDATA registry since 1978 (www.anzdata.org.au). Based on the ANZDATA registry, 508 adults and 19 children and young adults started RRT in New Zealand in 2015, with 503 starting dialysis, comprising an incidence rate of 115 pmp (Fig. 81.9).[9,96] The incidence of treated ESKD in New Zealand is similar to that of neighboring Australia (111 pmp in 2015) and has remained relatively unchanged over the last 15 years. The incidence of dialysis varies markedly by DHB catchment area with the highest rate for counties Manukau (including South Auckland; 611 pmp) and Waikato (496 pmp) and the lowest for Southern (45 pmp) and Canterbury (78 pmp) regions, with differences between regions related in part to ethnicity and socioeconomic opportunity.

Overall, 2674 people were treated with dialysis in New Zealand in 2015, accounting for a prevalence of 582 pmp. Together with kidney transplantation (369 pmp), the overall prevalence of RRT for Aotearoa/New Zealand in 2015 was 950 pmp, which is nearly identical to neighboring Australia (947 pmp), and comparable with the average of the nation states of the European Union (1090 pmp).

Older age groups have the highest incidence of RRT [311 pmp for the 65- to 74-year-old age group and 260 pmp for the 75- to 84-year-old age group (compared with 77 pmp for the 25- to 44-year-old age group)]. Children and young adults (0 to 25 years old) experienced an incidence of 13 pmp. The proportion of adults older than 85 years of age starting dialysis has fallen over time; zero patients over 85 years commenced dialysis in 2015.

The first RRT occurred in New Zealand in Wellington Hospital in 1954, performed by Dr. Neil Turnball and Dr. Dave Reid, in which PD was carried out.[97] The first hemodialysis was also performed in Wellington in 1958 by Dr. Verney Cable using a Kolff dialysis machine. The first home dialysis was done in 1968 in Auckland and the first home dialysis training program was established in Christchurch under the leadership of Dr. Peter Little.[98]

Overall, most prevalent patients with treated ESKD in New Zealand are treated with dialysis (61.2%; Fig. 81.9). New Zealand has the highest global rate of home-based dialysis therapy, although this is falling due to falling rates of PD (Fig. 81.10). In 2015, 18.0% of patients treated with dialysis in New Zealand did home hemodialysis, 29.6% did PD, and 52.5% received hospital hemodialysis.

In New Zealand, hemodialysis is accessed in three principal settings: facility (hospital or satellite), community house, or home.[99] Community house hemodialysis involves patients performing their hemodialysis independently without direct nursing or medical supervision in an unstaffed nonmedical community homelike setting, and is used in two DHB regions (Counties Manukau and Hawke's Bay), serving approximately 650,000 people. In a 10-year analysis between 2000 and 2010, 113 patients commenced dialysis in a community house setting.

New Zealand has the highest rate of home hemodialysis in the world (18.0%), which is similar to the rate in Australia (15.6%). The rate of home hemodialysis in New Zealand appears to have been stable over the last decade, despite a progressively more dependent dialysis population with increasing medical comorbidity. The principal secular change in dialysis modalities in New Zealand has been reducing utilization of PD (falling from 38.3% to 29.6% over 10 years) in favor of center-based hemodialysis (increasing from 45.4% to 52.5%; see Fig. 81.10). There is marked regional variation in home hemodialysis uptake, ranging from 12% of all dialysis patients treated at Capital and Coast District Health Board to 58% at the Southern District Health Board. Regional variation is likely related to policy (two DHBs, Canterbury and Southern, maintain a universal home dialysis program), remoteness, comorbidity, and socioeconomic factors. Home-based hemodialysis among children is rare.

As with modality practices, the hemodialysis prescription varies across regions related to the uptake and utilization of home-based therapy. The majority of New Zealand hemodialysis

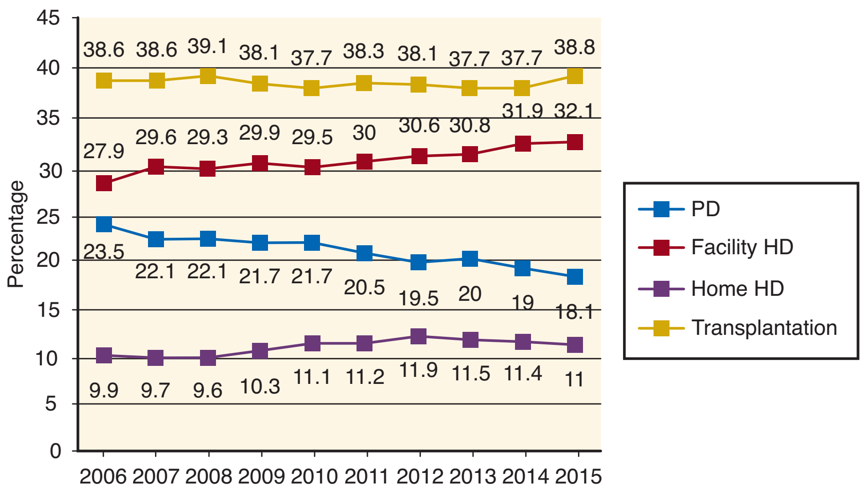

Fig. 81.9 Prevalence of renal replacement therapy modality in New Zealand 2006–2015. (From Ministry of Health National Renal Advisory Board. *New Zealand Nephrology Activity Report 2015*. Wellington, New Zealand: Ministry of Health; 2015. Available from: http://www.health.govt.nz/about-ministry/leadership-ministry/expert-groups/national-renal-advisory-board/papers-and-reports. Last accessed December 10, 2018.)

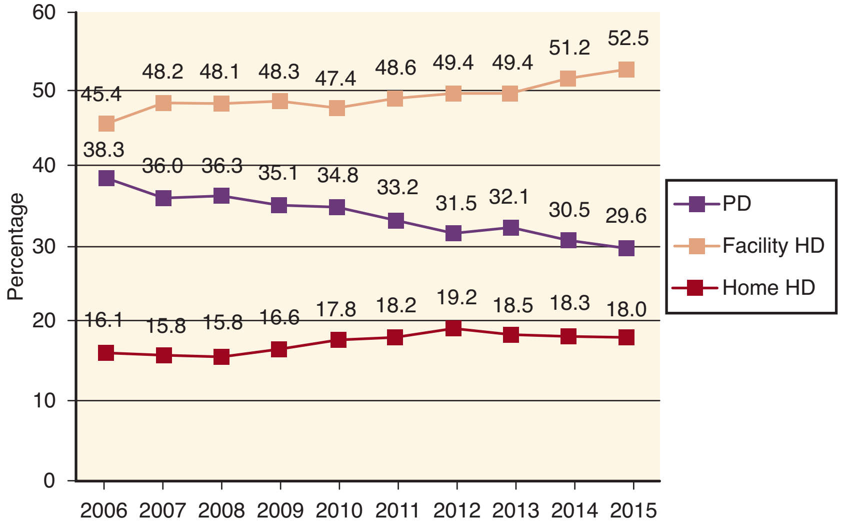

Fig. 81.10 Prevalence of dialysis modality in New Zealand 2006–2015. *HD,* Hemodialysis; *PD,* peritoneal dialysis. (From Ministry of Health National Renal Advisory Board. *New Zealand Nephrology Activity Report 2015*. Wellington, New Zealand: Ministry of Health; 2015. Available from: http://www.health.govt.nz/about-ministry/leadership-ministry/expert-groups/national-renal-advisory-board/papers-and-reports. Last accessed December 10, 2018.)

patients do hemodialysis 3 days a week (ranging from 51.6% of unit patients to >95%). At most centers, >90% of patients undergo three dialysis sessions each week. In centers with high uptake of home dialysis, 40%–50% of patients do dialysis either on alternate days or more frequently, while the national average is 18% for more frequent hemodialysis. The vast majority of New Zealand patients (>90%) undergo ≥4.5 hours of hemodialysis at each session. The most frequent blood flow rate used in New Zealand for patients with an arteriovenous fistula is 300–349 mL/min (59%), with smaller fractions prescribed 250–299 mL/min (16.7%) or 350–399 mL/min (16.6%). Hemodiafiltration practices have changed markedly since 2007, with a sharp increase from nearly zero to 20% in 2010, and with a subsequent plateau between 2010 and 2014. As the urea reduction ratio or Kt/V is not measured routinely at some centers, particularly those with a high prevalence of home dialysis, this measure of dialysis adequacy is not reliably ascertained for New Zealand hemodialysis patients.

About one-third of New Zealand patients commence hemodialysis with permanent arteriovenous vascular access (either fistula or graft), even when assessed by a nephrologist >3 months before starting RRT. About 80% of prevalent hemodialysis patients have permanent vascular access, although the prevalence decreases with age; approximately 40% of patients in the >75-year-old age group have a tunneled or nontunneled central venous catheter for dialysis access. The National Renal Advisory Board in New Zealand—the national leadership body advising on and monitoring renal care—has set the quality care standard for permanent vascular access at dialysis start (late referrals excluded) at 80% and prevalent permanent vascular access at 70%; however, these targets continue not to be achieved across the country.[100]

Median patient survival on hemodialysis in New Zealand varies with modality, age, and ethnicity.[101] The median survival for those aged 25 to 44 years is 8 years, 45 to 64 years is 5.5 years, 65 to 74 years is 3.7 years, 75 to 84 years is 2.8 years, and 85+ years is 2.2 years. There is a 52% lower risk of mortality with home dialysis during the first 3 years of dialysis and beyond, although this difference may not be present for Pacific patients.[102]

PD was utilized by 791 (30%) patients treated with dialysis in New Zealand in 2015. Unlike Australia (105 pmp), the prevalence of PD is decreasing, although at 172 pmp it remains higher than peer nations such as Germany (39 pmp), the United Kingdom (75 pmp), and the United States (87 pmp), likely driven by treatment policies in some regions (home dialysis) and geographical remoteness. Most patients (>50%) commencing PD are aged between 55 and 84 years, in parallel with the age groups with the highest incidence of treated ESKD (Fig. 81.11). In contrast to Australia, in which APD is utilized by 65% of those treated with PD, the use of APD in New Zealand is approximately 50%, at the upper end of the range of utilization reported among higher income countries (42.4%, 95% CI: 34.4 to 50.5).[29] Most (72%) of the 291 patients starting PD in New Zealand in 2015 were incident dialysis patients, while 27% transferred from hemodialysis, and 1% commenced PD after a failed transplant.

The median survival for PD in New Zealand is 3.92 years (95% CI: 2.11 to 6.34), although it varies substantially by age, comorbidity, and time.[102] PD is associated with a 20% lower mortality risk in the first 3 years of treatment in New Zealand, which may be offset by a 33% increased mortality during treatment >3 years. The survival benefit observed with PD may be lower among Māori and Pacific patients. Current 1-, 3-, and 5-year survival for PD is 92%, 63%, and 37% in those with diabetes, and 95%, 78%, and 63% in those without diabetes.[28]

While uptake of PD among incident patients has increased over the last 10 years in New Zealand (from 32% to 40% of all incident dialysis patients), the overall prevalence of the modality has been falling, hampered principally by technique survival and a consequent shift to hospital-based hemodialysis care.[103] Current estimates of technique survival for PD in New Zealand are 95%, 86%, 46%, and 16% at 6 months, 1,

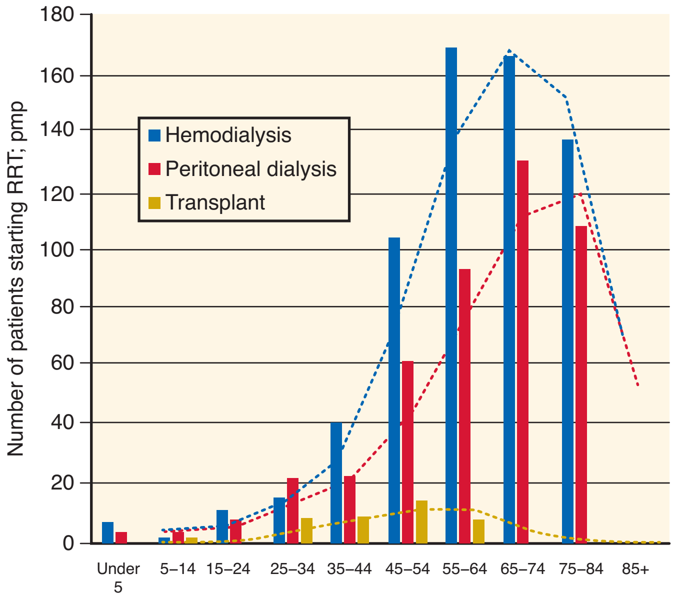

Fig. 81.11 Incidence of RRT by age group and modality in New Zealand, 2015. *pmp,* per million population; *RRT,* renal replacement therapy. (From Ministry of Health National Renal Advisory Board. *New Zealand Nephrology Activity Report 2015*. Wellington, New Zealand: Ministry of Health; 2015. Available from: http://www.health.govt.nz/about-ministry/leadership-ministry/expert-groups/national-renal-advisory-board/papers-and-reports. Last accessed December 10, 2018.)

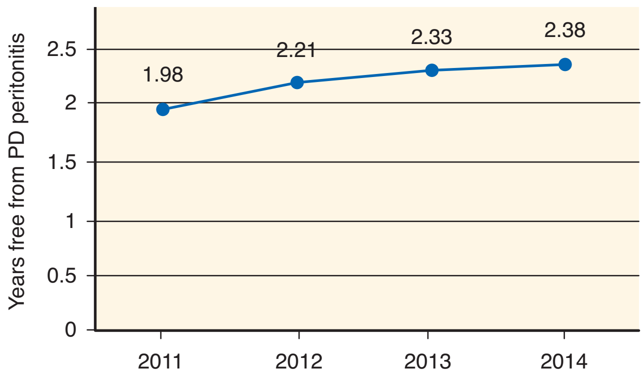

Fig. 81.12 National average years free from peritoneal dialysis (PD) infection in New Zealand between 2011 and 2014. *PD,* Peritoneal dialysis. (From Ministry of Health National Renal Advisory Board. *New Zealand Nephrology Activity Report 2015*. Wellington, New Zealand: Ministry of Health; 2015. Available from: http://www.health.govt.nz/about-ministry/leadership-ministry/expert-groups/national-renal-advisory-board/papers-and-reports. Last accessed December 10, 2018.)

3, and 5 years. Infection has been the most frequent cause of technique failure (44%), followed by failure of dialysis function (23%), technical issues (19%), and change in modality due to patient preference (7%). Unlike in Australia, in which the rate of PD-related peritonitis has fallen since 2009 as a result of several quality improvement initiatives,[20] the rate of PD peritonitis in New Zealand has improved at a smaller rate over time (Fig. 81.12) and remains somewhat higher than for Australia.

The costs of dialysis are not publicly available. Estimations of dialysis and transplant treatment costs are available from the Auckland Regional Renal Report in 2002–2004. In that report, annual costs (including routine dialysis treatments, hospitalizations, physician assessment not related to dialysis, and transportation) were estimated at US$44,053 for hospital hemodialysis, US$32,995 for satellite hemodialysis, US$25,078 for continuous APD, and US$23,003 for home hemodialysis.[95] Costs of APD are not available.

TRANSPLANTATION

Three centers perform kidney transplantation in New Zealand: Auckland, Wellington, and Christchurch. Simultaneous kidney and pancreas transplants are performed in Auckland, as are combined solid organ transplants. The first organ to be transplanted in New Zealand was a kidney at Auckland Hospital on May 28, 1965, about 3 months after Australia's first kidney transplant. Lung transplantation started in 1993 and liver and pancreas transplantation in 1998. Organ Donation New Zealand arose from an expanded and renamed national Transplant Donor Coordination Office in 2005 and is the national deceased-organ donation agency. In 2012, the New Zealand Government provided NZ$4 million for initiatives to increase living and deceased donation rates by funding Organ Donation New Zealand, the Counties Manukau District Health Board "Live Kidney Donation Aotearoa" project, and the Auckland District Health Board to formalize the Kidney Exchange Program. An additional NZ$4 million was allocated in 2014 for the establishment of the National Renal Transplant Service to increase and improve transplantation processes, increase donor liaison coordinator capacity, and provide further funding for the New Zealand Kidney Exchange program (NZKX).

In 2015, there were 147 kidney transplants in New Zealand, which is the highest number ever performed in 1 year in the country (Fig. 81.13), and represents a rate of 32 pmp. The number of transplant operations each year has sharply increased in the last 2 years and is likely to be related to the quality improvement initiatives arising from government funding over the previous decade.[104]

The transplantation rate (32 pmp) remains substantially lower than the overall incidence of ESKD (115 pmp), and somewhat lower than the transplantation rate for Australia (40 pmp). The number of kidney transplants each year in New Zealand (147 in 2015) represents 31% of the 472 patients who are active on the kidney transplantation waiting list. Unlike Australia, in which most transplants were from deceased donors (70%), 50% of kidney donors in New Zealand in 2015 were deceased. In 2015, there were 19 preemptive kidney transplants, representing 4.6% of all patients starting RRT. The preemptive transplantation rate has changed little over the last decade but does vary by ethnicity (see Fig. 81.8).

As in Australia, recent developments in New Zealand transplantation services have included the NZKX and ABO-incompatible kidney transplantation. In 2015, there were two kidney exchange chains providing four kidney transplants and nine ABO blood group–incompatible kidney transplants were performed.

During the last 5 years and consequent to increasing transplantation rates, the prevalence of patients living with a functioning kidney transplant has increased in New Zealand from 1483 patients (338 pmp) to 1694 patients (369 pmp). There continues to be marked variation between DHBs in the overall prevalence of kidney transplantation, ranging from 238 pmp to 508 pmp. This variation reflects local transplantation practices, patient comorbidity, and the underlying rates of ESKD. Transplantation most commonly occurred in the 55–64-year-old age group, consistent with a higher incidence of ESKD at this age. As a proportion of all patients with treated ESKD, the age group with the highest prevalence of transplantation was the 5–14-year age group. Over 90% of children and adolescents live with a functioning transplant, with most receiving a kidney transplant from a living donor.

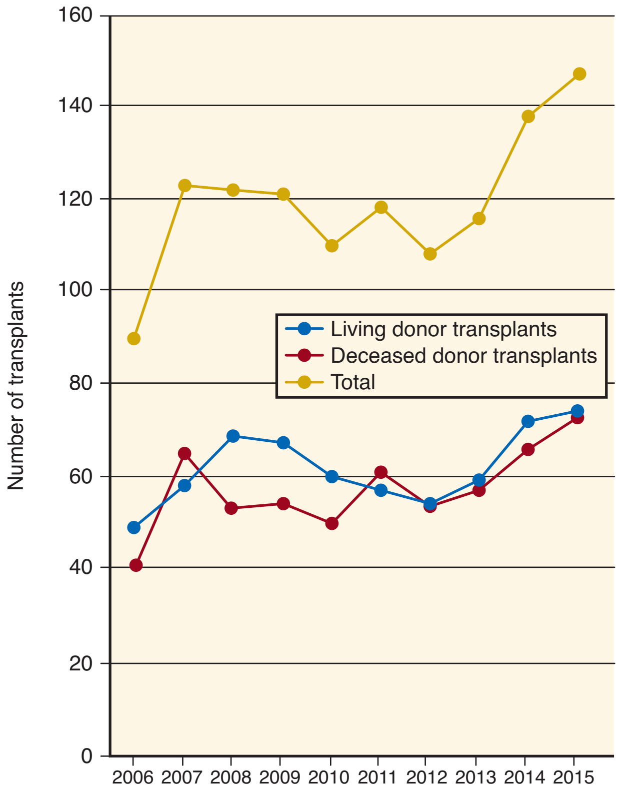

Fig. 81.13 Living and Deceased donor kidney transplantation in New Zealand during 2006 and 2015. (From Ministry of Health National Renal Advisory Board. *New Zealand Nephrology Activity Report 2015*. Wellington, New Zealand: Ministry of Health; 2015. Available from: http://www.health.govt.nz/about-ministry/leadership-ministry/expert-groups/national-renal-advisory-board/papers-and-reports. Last accessed December 10, 2018.)

Graft survival rates in New Zealand for first grafts are presented in Table 81.2. The 1-year patient and graft survivals have remained stable across recent transplantation eras (2008–2015) for both recipients of a deceased-donor transplants and living-donor transplants. The patient and graft survivals remain approximately 90% and 80%–85%, respectively, for living- and deceased-donor transplant recipients in New Zealand, although data are susceptible to variation due to the relatively small numbers of transplants performed.

As is observed in Australia, the most frequent cause of graft loss is death with a functioning transplant (185 patients during 2011–2015; 51.3%), followed by chronic allograft nephropathy (113 patients; 31.3%). The most common causes of death for transplant recipients are cancer (59 patients accounting for 32% of all deaths) and CVD (also accounting for 32% of all deaths). Increased risks of cancer are observed at numerous body sites after kidney transplantation, principally representing cancers of known or suspected viral etiology, suggesting a role for impaired immunological surveillance.[105]

NONDIALYSIS SUPPORTIVE CARE FOR ESKD

Increasing recognition of nondialysis care for ESKD has led to the emergence of nondialysis supportive care services within hospital-based nephrology care in New Zealand, as has been similarly observed in many higher-income countries. Nondialysis supportive care can offer patients information and support when considering whether or not to commence dialysis or withdrawing from dialysis care, and for management of symptoms related to ESKD. Supportive care services in New Zealand are not coordinated at a national level and represent local care initiatives related to existing capacity and expertise.[100] Supportive care practices frequently include multidisciplinary teams including palliative care and community nursing services. For many patients in primary care, palliative management of ESKD is coordinated within general practice without the involvement of secondary nephrology services.

ACUTE KIDNEY INJURY

Few systematic data exist for the incidence and outcomes related to AKI in Aotearoa/New Zealand in adults. AKI occurred in 5.2% of patients at the time of or during the first 24 hours of intensive care unit admission, with a rising

Table 81.2 Patient and Graft Survival for First Grafts in Aotearoa/New Zealand by Donor Type for Transplants Occurring in 2010–2011

Outcome	1 Month	6 Months	1 Year	5 Years
Patient Survival				
Deceased donor (n = 103)	99 (93–100)	99 (93–100)	98 (92–100)	88 (79–93)
Living donor (n = 111)	100	99 (94–100)	97 (92–99)	90 (82–94)
Graft Survival				
Deceased donor (n = 103)	96 (90–99)	95 (89–98)	94 (87–97)	82 (73–89)
Living donor (n = 111)	97 (92–99)	96 (91–99)	95 (88–98)	81 (72–87)

Data are shown with 95% confidence interval.

incidence noted during the years of 1996 to 2005. Incidence increases with age and comorbidity, and is more frequent for patients with liver disease (12.1%) or hematological malignancy (10.5%) admitted to intensive care. Crude hospital mortality for patients admitted to intensive care with AKI (42.7%) is substantially higher than for those patients without AKI (13.4%), although in the decade through to 2006, mortality decreased among patients with AKI.[106] AKI is associated with a hospital stay of 19.8 days on average for those patients surviving to discharge, and patients with AKI are more likely to be transferred to further acute care or rehabilitation.

For children, the annual incidence of AKI between 2001 and 2006 was 4.0 per 100,000 children under 15 years of age, most commonly associated with cardiac surgery (58%), hemolytic uremic syndrome (17%), and sepsis (13%).[107] Survival of AKI to hospital discharge for children was 89%, with evidence of ongoing kidney impairment in 40%. Māori and Pacific children are treated for AKI more often than New Zealand European children.

KIDNEY DISEASE CARE IN NEW ZEALAND

HEALTH FINANCE

Before World War II, the health care in New Zealand was predominantly privately funded, with around 57% of total health funding from the private sector in 1925.[92] Public contributions to health funding increased, such that in 1945, 74% of health expenditure was from public sources, with a peak in the 1980s of 88%. Direct funding from the government, through the Ministry of Health, is the largest source of funding of the health and disability sector.

In 2000, the New Zealand Public Health and Disability Act allowed the implementation of DHBs. There are currently 20 DHBs in New Zealand that are responsible for planning, and providing or purchasing health and disability services within their geographical region, including acute hospital services.

CKD and AKI are managed collaboratively across the health sector in primary, secondary, and tertiary care. At present, hospital-based dialysis services are provided by 11 DHBs to their own regions and other DHBs. CKD is most commonly identified within primary care settings, and referral is made to nephrology services for virtual or in-person consultation.

New Zealand's total health expenditure in 2016 was NZ$15.6 billion, representing approximately 9.5% of GDP in 2013.[108] Growth in health spending slowed following the global financial crisis and greater use of pharmaceutical spending on generic medications (accounting for 34% of pharmaceuticals in 2013). New Zealand ranked 12th in OECD nations for spending on health as a share of GDP in 2013.

Total expenditure on ESKD care is about 1% of total health expenditure. ESKD treatment is nearly entirely funded through the tax-funded health system. All inpatient and outpatient secondary-level health-care services including hospital-based care are free to patients, as are pharmaceuticals prescribed within secondary services. Primary care services are subsidized, although patients may be required to pay for services above any subsidy they receive. Similarly, prescriptions in the primary care setting accrue direct costs to the patient.

DHBs are provided a budget weighted according to the services they provide and the demographic and socioeconomic characteristics of their population. Budgetary constraints occur, such that DHBs are incentivized to manage funds within their allocation including dialysis and transplantation care. Private insurance can enable patients to recoup costs for pharmaceuticals and consultation within primary care, but cannot be applied to dialysis or transplantation care. A recent law change resulted in the Compensation for Live Organ Donors Act 2016, which provides financial assistance for lost income and childcare assistance for 12 weeks after living donation of kidney or liver tissue.

ESSENTIAL MEDICATIONS AND TECHNOLOGY ACCESS

Pharmaceutical Management Agency (Pharmac) is the government agency that determines which pharmaceuticals are publicly funded in New Zealand. Pharmac manages the Combined Pharmaceutical Budget which is set each year by the Minister of Health, and includes funding for community medicines, some hospital medicines, and some devices. The funding decisions of Pharmac are listed in the Pharmaceutical Schedule that defines subsidies for specific medicines and devices. A copayment for pharmaceutical items is set by the government and is free for children under the age of 13 years and about NZ$5 per item for the first 20 items prescribed to a family each year.

Many New Zealand patients live considerable distances from dialysis and transplant units, and for children, there is a single hospital in Auckland that provides in-patient and outpatient pediatric nephrology care in New Zealand. A National Travel Assistance program provides funding for transport and accommodation for patients required to travel and live in regions distant from their home for specialist care. Qualitative research has documented the considerable social disruption arising from dialysis treatment in New Zealand because of living remotely from dialysis and transplantation care.[109]

DETECTION AND MONITORING OF KIDNEY DISEASE

Late assessment by a specialist nephrologist (within 3 months of requiring RRT) is associated with poorer clinical outcomes and reduces the likelihood of preemptive transplantation.[110] Timely assessment for RRT requires detection of progressive kidney disease particularly within primary care practice, and collaborative management between primary and secondary care. In New Zealand, approximately 13% of patients experience late assessment, although this proportion is decreasing.[90] Younger adults and children are at particularly higher risk of late referral to specialist care; 26% of children and adults younger than 25 years old were referred within 3 months of RRT in 2015. Early detection of kidney disease and appropriate referral of patients who are likely to progress to ESKD may improve clinical outcomes, although empirical data suggest that large-scale implementation of global referral guidelines (Kidney Disease Improving Global Outcomes guidelines) may not be feasible under usual clinical practice models.[111]

There have been some nationwide programs in New Zealand to guide primary and secondary care practitioners in assessment and referral of patients with CKD. In 2015, a national consensus statement on managing CKD in primary care was released by the Ministry of Health[112] intended to articulate best practice in identification and management of CKD in primary care. This statement included systematic screening in primary care linked to existing diabetes and CVD screening programs, management of cardiovascular risk, and enablers of monitoring and recall systems for CKD care.

Strategies to support primary care of CKD include HealthPathways, which are agreements between primary and specialist services on how patients with specific health conditions will be managed within the local health service context. HealthPathways are principally designed to support primary care clinicians to manage patients in the community and guide referral practices for specialist assessment (www.healthpathwayscommunity.org). Decision aids for patients and clinicians making choices about treatment for ESKD have been developed by Kidney Health New Zealand an established patient advocacy group.[113]

A randomized clinical trial of community-based visits by nurse-led health-care assistants for medication adherence and algorithm-guided blood pressure care in Māori and Pacific patients led to greater use of medications, lower blood pressure, and reductions in proteinuria, suggesting an effective model of primary care for CKD.[114] Challenges to wider adoption of newer primary care practices to prevent ESKD include limited resourcing for nursing, availability of secondary care services to upskill primary care clinicians, and adequate protocols to integrate medical management across primary and secondary care. A recent systematic review of studies examining the effectiveness of chronic disease management programs of indigenous people in Australia, New Zealand, and Canada demonstrated that effective and acceptable programs were those that were integrated in existing health services, were nurse led, provided intensive follow-up, had governance structures supporting local leadership, included robust clinical systems for communication, and involved indigenous health workers as central to practice.[115] Use of telehealth within secondary nephrology services in New Zealand is not systematic. Telehealth-based services for long-term conditions in Australia and New Zealand lack high-level evidence of effectiveness.[116]

HEALTH WORKFORCE

At present, there are 423 medical graduates each year in New Zealand. Increased funding for medical education has led to a forecast expansion in numbers, which are expected to plateau in 2022 at 553 each year.[117] Reductions in international medical graduates suggest that additional training opportunities for New Zealand graduates within the health workforce will increase.

Recent workforce estimates in New Zealand indicate there are 230 full-time equivalent nurses providing care in hospital dialysis units and 21.5 full-time-equivalent nurses in home hemodialysis roles (R. Walker, personal communication). It was estimated in 2009 that there was one nurse for every 6.3 patients compared with 1 nurse for every 4.2 patients in Australia.[118] Nurses supporting CKD services specifically within secondary services are rare, with 2.5 full-time equivalents in hospital care in New Zealand. There are currently 6.1 full-time equivalent transplant coordinators and 3.9 full-time pharmacist equivalents.

In 2015, there were 45.4 full-time nephrologists (11.3 pmp), equivalent to 1 nephrologist for every 96 patients with ESKD (similar to 1 specialized nephrologist per 102 patients with ESKD reported in March 2004) and compares with 1 specialized nephrologist for every 80 patients with ESKD in Australia.[95] Historically, there have been insufficient New Zealand-trained nephrologists for workforce requirements, such that nephrologists have received prioritization within immigration processes. The nursing workforce in New Zealand is anticipated to be insufficient for an aging population, attributed to an older nursing workforce, reduced arrival of overseas-trained nurses, and increasing complexity of health services with multimorbidity.[119] The existing participation of Māori and Pacific practitioners in the New Zealand health workforce is low. In 2009, Māori and Pacific clinicians represented 3.1% and 1.5% of medical practitioners and 6.3% and 1.8% of the nursing workforce, respectively.[120] Specific programs have been established to increase Māori and Pacific participation in the medical workforce, which is considered to be a factor in reducing inequitable outcomes for Māori and Pacific communities.[121]

HEALTH INFORMATION SYSTEMS AND STATISTICS

Health-related information for patients with treated ESKD has been systematically collated in the ANZDATA Registry since 1978. Data about the incidence and prevalence of dialysis and transplantation treatment, together with information about treatment practices and clinical outcomes, are reported annually and as individual reports to DHBs. While comprehensive, ANZDATA does not include health information about untreated ESKD or earlier stages of CKD.

New Zealand has a National Health Index that assigns a unique number to each person in New Zealand who receives health care. This includes information on name, address, date and place of birth, gender, resident and citizen status, ethnicity, and (if appropriate) date of death.[122] The National Health Index number can be linked to other national and international health information repositories. An important aspect of health information used to drive health-care improvement is the accurate recording of ethnicity data. Specific protocols for the collection, recording, and output of ethnicity data are described for the Health and Disability Sector, with a specific focus to improve health outcomes and reduce health inequity.[123] Key requirements include ethnicity as self-identification, ethnicity is context specific and can change, that more than one ethnicity can be recorded, and that a standardized approach increases the accuracy of ethnicity data for decision-making.

At present, the number of people in New Zealand with CKD is unknown, as these data are not systematically recorded. Similarly, there is no existing framework for measurement of AKI or untreated ESKD in New Zealand. It is likely, due to increasing availability of large data sets, routine capture of diagnostic codes, and the National Health Index, that additional census data for kidney diseases will become available in the future.

STRATEGIES AND POLICY FRAMEWORKS

The New Zealand Ministry of Health has a Long-Term Conditions Program that provides support to the Ministry, health sector, and nongovernmental organizations to provide leadership, evaluate progress, and support clinical quality initiatives.[124] CKD is included in this framework as a condition that leads to an ongoing, long-term, and recurring impact on well-being. A National Health Strategy, released in 2016, has outlined higher-level direction and actions to improve the New Zealand health system between 2016 and 2026.[125]

Specifically, for CKD, the Ministry of Health provides leadership through the National Renal Advisory Board comprising clinical and managerial leaders in DHBs and professional and consumer groups.[100] The board holds responsibility for national strategy, projecting treatment demand, and defining service specifications for conditions involving the renal and urinary tract, and reporting the health activities and outcomes in nephrology services to the Ministry of Health. As discussed earlier in this chapter, the National Renal Transplant Service was established in 2014, and is implementing quality improvement metrics for transplantation. Quality improvement metrics include 1. the proportion of donor–recipient pairs who undergo transplantation surgery should receive surgery within 4 months of being accepted by the transplant service and are ready to proceed; and 2. the proportion of kidney transplants from living donors occurring as preemptive or early kidney transplants. In addition, the New Zealand Kidney Allocation Scheme was developed in 2016 as an algorithm for allocation of deceased-donor kidneys and nondirected live-donor kidneys to maximize equity, accountability, and transparency.[126]

In addition to national bodies and practice frameworks, clinical practice guidelines for care of dialysis, transplant, and earlier stages of CKD are provided through the KHA-CARI.

CAPACITY FOR RESEARCH AND DEVELOPMENT

Health research in New Zealand is predominantly conducted within tertiary education institutions and DHBs. New Zealand spent 1.2% of GDP on research and development in 2014, which is substantially lower than the average among OECD countries of 2.4%. The New Zealand Government spending on research was 0.27% of GDP in the same period, and represents considerably lower investment than in other small advanced economies, such as Israel (>4%), Denmark and Finland (>3%), and Singapore (>2%).[127] A New Zealand Health Research Strategy developed by the Ministry of Health and the Ministry of Business, Employment and Innovation, together with the primary governmental health research funder, the Health Research Council, has recently been released, and is designed to increase implementation of health research findings and retain clinicians with an interest in health research.[128]

There is limited coordination in research for kidney disease in New Zealand with a range of researchers and small research aggregates drawing funding from private and public sources. The AKTN is a nonprofit collaborative research group that supports the collaborative development of trials in kidney disease in both Australia and New Zealand and which supports funding applications for trial conduct. The New Zealand Institute for Pacific Research is aimed to promote and support excellence in Pacific Research, although to date there are no specific collaborative research projects with Pacific and New Zealand researchers specifically for kidney disease.[129]

SAMOA

Samoa is made up of six islands, and is a member of the Commonwealth of Nations. From the end of World War I until 1962 when Samoa attained independence, it was under the control of New Zealand. Similar to other island nations in the Pacific, diabetes is highly prevalent in Samoa, and CKD is listed within the top 10 causes of deaths in the country (ranked fourth in 2015).[130] Samoa's health system is made up of public and private health sectors and the first hemodialysis unit (12 dialysis machines) opened in 2002 with assistance from the Singapore National Kidney Foundation.[131] There are currently two centers located in two main islands (Upolu and Savaii), providing treatment for 88 patients who are on a waiting list to access the service. The government of Samoa has acknowledged the strain imposed by the growing burden of CKD, and is currently working on public awareness campaigns and evaluating methods to promote healthier lifestyle measures to lower risk factors for CKD, such as hypertension and diabetes.[132]

NEW ZEALAND DEPENDENCIES (COOK ISLANDS, NIUE, TOKELAU)

Cook Islands, Niue, and Tokelau are all self-covering island countries in the South Pacific Ocean in free association with New Zealand. Cook Islands is the most populous of the group, which comprises 15 islands. Tourism is the country's main industry as it lacks major natural resources (makes up approximately 67.5% of GDP).[133] Although Cook Islands is thought to perform better than other Pacific island nations in relation to health indicators, there has been a 40%–50% increase in the prevalence of diabetes followed by increasing number of CKD patients (expected to more than double in the next 10 years).[134] Noncommunicable diseases such as CKD are reported to be the leading causes of mortality in the Cook Islands (responsible for 75% of deaths).[135] According to a report generated in 2012, there were 26 physicians working in Cook Islands, but whether this includes a nephrologist is uncertain.[136] Patients with ESKD, as New Zealand citizens, receive dialysis treatment within the New Zealand tax-funded public health system presently, as there is no capacity for RRT within the Cook Islands. The majority of patients receive hospital-based hemodialysis in New Zealand, driven by patient preference.[131]

Niue is located 2400 km northeast of New Zealand and west of Cook Islands, with the population made up predominantly of Polynesians (around 1612 as of November 2016). Over 90% of Niuean people live in New Zealand, and Niue was the first country in the world to offer free wireless Internet to all its inhabitants.[137] Similarly, about 1400 people are thought to reside in Tokelau, which is another island country located north of the Samoan Islands. It is known as the first 100% solar-powered nation in the world.[138] Given their small population with limited data available on health, there is no known information pertaining to the burden of AKI or CKD in either of these two nations.

PAPUA NEW GUINEA

Papua New Guinea is classified as a low- to middle-income country with close to 90% of the population living in rural areas. Primary health-care facilities are the main point of access to the health system, but access to these services faces many geographic, cultural, financial, and systemic challenges.[139] Many villages can only be reached via air or foot, where over 90% of existing roads are unpaved.[140] Over the past 20 years, there has been a deterioration in the availability of primary health-care services,[141] with up to 40% of aid posts closing, thereby restricting access to health care. Moreover, in spite of a predominant public funding base supplemented in some areas with church-run facilities supported by grants from the National Department of Health, many facilities charge a user fee, which creates further barriers to health care for those who cannot afford the service.[142]

The exact prevalence of CKD in Papua New Guinea is unknown, however CKD was ranked as the eighth most common cause of death in 2015.[143] Increase in the burden of CKD is likely to be driven by growth in prevalence of diabetes mellitus, which is thought to affect 12.9% of the adult population in Papua New Guinea.[144] This is concerning, as access to RRT is inadequate. Although PD is available, a limited number of hemodialysis machines are available (reported to be two in 2014),[145] with no capacity for renal transplantation.

PACIFIC ISLANDS

FRENCH PROTECTORATES (FRENCH POLYNESIA, NEW CALEDONIA, WALLIS AND FUTUNA)

French Polynesia, New Caledonia, Wallis and Futuna form French protectorates within Oceania. French Polynesia is made up of 118 islands scattered over 2000 km of ocean, divided into five groups of islands, with 67 islands that are inhabited.[146] Tahiti is the most populated island of the group and serves as the capital of the collectivity. Although the Government of France directs justice, university education, security, and defense, the local government controls health.[146] French Polynesia has a moderately developed economy with modest GDP per capita amongst countries within Oceania (Table 81.1). Health facilities are considered to be of reasonable standard, however access to medical services, including dialysis, is restricted in less-populated islands. There is only one hemodialysis unit, located in Tahiti.[147] The medical practitioners operate in private practice, with consultation costs up to 3500 Comptoirs Français du Pacifique Franc to see a GP with a higher cost for a specialist review.[148] Diabetic nephropathy is a major cause of CKD in this region,[149] with relatively high background prevalence of diabetes mellitus (7.3%).[150] Another important cause of ESKD in French Polynesia is Alport syndrome, which affects 18% of the patients receiving dialysis[151] (compared with 1%–2% of ESKD patients in Europe, and 2.3% of renal transplant patients in the United States[152]). It is the most prevalent monogenic inherited disorder found within French Polynesia.[151]

New Caledonia is another special collectivity of France, located within the Melanesia subregion, formed by several islands, including the main island of Grand Terre. It has one of the largest economies in the South Pacific, with GDP per capita similar to New Zealand (Table 81.1). According to a retrospective renal biopsy study, which included 202 renal biopsies from 181 patients in New Caledonia, the most prevalent primary glomerular disease was focal segmental glomerulosclerosis and the most prevalent systemic glomerulopathy was amyloidosis. Other common pathological findings included postinfectious glomerulonephritis, minimal change disease, and diabetic nephropathy.[153] In 2009, the most common conditions requiring long-term treatment included diabetes (9509 cases, 17.6%) and 1198 cases of renal failure (2.2%) requiring treatment. Given the growing burden of diabetes in the community, CKD from diabetic nephropathy is likely to have also increased over time. The local government has endorsed the "Health for All" principle, with provision of public and private health-care services (with 2.9 hospital beds available for every 1000 people). Economic growth of New Caledonia and improvement in quality of health-care coverage have led to unfettered access to health services by the whole population.[154] Access to both hemodialysis and PD is available.[147]

The Territory of the Wallis and Futuna Islands is made up of three main islands with a smaller number of tiny islets. The vast majority of the population is of Polynesian ethnicity, with a small minority of French descent. A large proportion of Wallisians and Futunians live as expatriates in New Caledonia due to limited resources on their homelands, which has led to a decreasing trend in population over time (14,166 in 1996 vs. 12,197 in 2013).[155] The Health Agency for the Territory of the Wallis and Futuna Islands is a French national public administrative agency with the aim to ensure health protection in the Territory. There is a high prevalence of risk factors for chronic noncommunicable diseases, including cigarette smoking (49.1%), obesity (59.5%), hypertension (34.1%), diabetes mellitus (17.1%), and dyslipidemia (16.1%) according to a study, which comprised 560 people from the Territory (representing 6.71% of the adult population living in Wallis and Futuna).[156] In the studied population, moderate CKD, defined as an eGFR of 30–59 mL/min/1.73 m^2, was present in 5.8% of the participants.[156] At present, there is no dialysis unit in the Territory.

US PROTECTORATES (AMERICAN SAMOA, GUAM)

American Samoa is an unincorporated territory of the United States consisting of five main islands. It has a small developing economy with two main sources of income, the United States Government and tuna canning, which make up 93% of the economy together.[157] The most significant health problems relate to the increase in chronic diseases associated with lifestyle, due to poor nutrition and physical inactivity, such as hypertension, CVD, and type 2 diabetes mellitus.[158] According to the American Samoa noncommunicable STEPwise approach to Surveillance (STEPS) survey, conducted between June and August 2004, 74.6% of American Samoans were obese (defined as BMI $\geq$ 30 kg/m^2) and 93.5% were classified as overweight (BMI 25–29.9 kg/m^2) or obese, whilst hypertension and diabetes were present in 34.2% and 47.3% of the adult population, respectively.[159] Although the exact burden of CKD is uncertain, it is likely to be significant and growing in the context of increasing burden of risk factors for CKD. There is one hemodialysis unit at the Lyndon B. Johnson Tropical Medical Center in Pago Pago, which has 16 dialysis chairs,[159] where 160 patients with ESKD receive hemodialysis treatment each week.[160] The number of patients receiving hemodialysis in American Samoa has increased by 33% from 2006 to 2010.[159]

Guam is another unincorporated and organized territory of the United States. However, unlike American Samoa, Guamanians are American citizens by birth. Guam has a land area of 544 km^2 (75% the size of Singapore), and is the largest island in Micronesia. The largest ethnic group is the native Chamorros (37.3%), followed by Filipinio (26.3%), White (7.1%), and Chuukese (7%) ethnicities. Guam's economy depends primarily on tourism.[161] Similar to other Pacific Island nations, the most common cause of ESKD is diabetic nephropathy, particularly amongst Pacific Islanders.[162] US board–certified medical practitioners serve the people of Guam in all specialties and health care is available through public and private sectors.[161] There were four dialysis units and six nephrologists in 2014,[163] and there is no transplant program in place.

NORFOLK ISLAND (AUSTRALIAN PROTECTORATE)

Norfolk Island is located between Australia, New Zealand, and New Caledonia, which forms part of the Commonwealth of Australia. Given close ties with Australia and New Zealand, with a small economy, there has been a growth in emigration that has led to a decreasing trend of population (1796 people in 2011 compared with 2601 in 2001).[164] Norfolk Island Hospital is the only medical center on the island with three medical practitioners.[165] Medicare and PBS cover medical treatments, analogous to mainland Australia. However, the hospital is not equipped to treat any serious emergencies. In situations where health conditions deteriorate in residents of Norfolk Island, many are forced to leave their homes and relocate to New Zealand or Australia to receive medical care, including RRT.[164]

PITCAIRN ISLAND (BRITISH COLONY)

The Pitcairn Islands is the last British Overseas Territory of the Pacific. Of the four islands (Pitcairn, Henderson, Ducie, and Oeno), only Pitcairn is inhabited by approximately 50 permanent people including one government-employed medical practitioner, and is the least populous national jurisdiction in the world.[166] The island can only be accessed by yachts or ships and receives a scheduled supply boat at 3-monthly intervals.[167] Its remote location with a lack of access to services including hospital and reliable power has been blamed as a major barrier to growth. Like other island nations of the Pacific, diabetes mellitus is reported to be highly prevalent,[168] but the burden of CKD is unknown.

FIJI

Fiji is an island country in Melanesia formed by >330 islands, of which 110 are permanently inhabited. Over 80% of the population resides on the two main islands: Viti Levu and Vanua Levu. Fiji has one of the most developed economies in the Pacific due to an abundance of natural resources and an active tourism sector. CKD has been identified as the fourth most common cause of death in 2015 (increase by 34.0% since 2005).[169] A recent STEPS survey has reported 30% of Fiji's adult population as diabetic and the incidence of stage 5 CKD is estimated to be the highest in the world, at around 680 pmp.[170] Fiji suffers from an underdeveloped health system, and there is only one formally trained nephrologist in Fiji (1.1 pmp). This has led to difficulties to screen and treat early-stage CKD and the urgent need to address noncommunicable diseases including CKD has been recognized by the government. Although chronic hemodialysis is available in three units (Suva, Labasa, and Nadi), it is only accessible to those who are able to afford the service, and long-term treatment is rare due to financial constraints. In 2016, there were 107 dialysis patients in Fiji. PD and kidney transplant programs are currently unavailable in Fiji. Living-donor kidney transplants are performed in India.[170] There are plans to expand capacity to support patients with ESKD in Fiji, especially those who could not previously afford RRT, with recent government announcement of allocating $1 million for the establishment of the National Kidney Research and Treatment Centre and an additional $300,000 for dialysis treatment.[171] In addition to three chronic hemodialysis units, two of the three public hospitals are able to perform dialysis for treatment of AKI (A. Krishnan, personal communication). One of the important causes of AKI in Fiji has included leptospirosis, which is identified as one of the four priority climate-sensitive diseases of major public health concern.[172] Outbreaks of leptospirosis have frequently occurred in the setting of floods, which resulted in 576 reported cases and 40 deaths (7% case fatality) in 2012.[173] Some of the identified risk factors have included male gender, indigenous Fijian status, young adults (15 to 45 years of age), living in rural areas, and abattoir workers, with peak seasons following rainy months and cyclones.[174,175]

CONCLUSION

Oceania is a region which comprises >10,000 islands with an estimated population of >40 million people with a large variation in the level of economic status. The burden of CKD is increasing across the region, driven by inexorable growth in noncommunicable chronic illnesses, such as diabetes mellitus. Access to health care and data on health outcomes or burden of disease are highly variable and mostly very limited in many nations of the Oceania region outside of Australia and New Zealand. Even within affluent countries, such as Australia and New Zealand, there exist inequities in health outcomes affecting indigenous patients. In response to growth in the prevalence of CKD and its associated adverse health outcomes, various governments are currently expanding their capacities to improve care for these patients. Implementing infrastructure to capture the true disease burden and health outcomes will be an important component of strategies to achieve better health outcomes.

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BOARD REVIEW QUESTIONS

1. With respect to kidney disease in Oceania, which of the following is TRUE:
   a. Chronic glomerulonephritis is the commonest cause of chronic kidney disease in Oceania.
   b. Alport syndrome is the most common monogenic inherited disorder in French Polynesia.
   c. IgA nephropathy is the commonest form of primary glomerular disease in New Caledonia.
   d. The prevalence of diabetic kidney disease in the Pacific Islands is lower than that in Australia and New Zealand.
   e. Leptospirosis is an uncommon cause of acute kidney injury in Fiji.

   Answer: b

   Rationale: Alport syndrome is the most prevalent monogenic inherited disorder found within French Polynesia, accounting for 18% of the patients receiving dialysis (compared with 1%–2% of end-stage kidney disease patients in Europe, and 2.3% of renal transplant patients in the United States). Diabetic kidney disease is the commonest cause of chronic kidney disease in Oceania and is more prevalent in the Pacific Islands than in Australia and New Zealand. The prevalence of diabetic kidney disease in Fiji is one of the highest in the world. The most prevalent primary glomerular disease in New Caledonia is focal segmental glomerulosclerosis. Leptospirosis is a common cause of acute kidney injury in Fiji and most commonly affects young adult males, particularly following rainy periods.

2. Which of the following statements about renal replacement therapy in Oceania is FALSE:
   a. Australia has the highest rate of home hemodialysis in the world.
   b. Kidney transplantation is generally not available in the Pacific Islands.
   c. The prevalence of treated end-stage kidney disease in Australia and New Zealand is less than half that of the United States.
   d. Approximately 50% of people who die from end-stage kidney disease in Australia do so without accessing renal replacement therapy.
   e. Hemodialysis session durations in Australia and New Zealand are on average more than an hour longer than those in the United States.

   Answer: a

   Rationale: New Zealand has the highest rate of home hemodialysis in the world. Australia has the second highest rate of home hemodialysis in the world. All of the other statements are true.

3. Which of the following statements about ethnic inequalities in Australia is FALSE:
   a. The proportion of indigenous Australians with end-stage kidney disease due to diabetic kidney disease is approximately double that of nonindigenous Australians.
   b. The hospitalization rate for regular dialysis treatment in indigenous Australians is 11 times as high as for other Australians.
   c. Hospitalization rates for acute kidney injury in indigenous Australians are almost three times higher than those of nonindigenous Australians.
   d. Indigenous Australians are less likely to receive a kidney transplant than nonindigenous Australians.
   e. Indigenous Australians living in urban locations have a higher risk of developing end-stage kidney disease than those living in rural and remote locations.

   Answer: e

   Rationale: Indigenous Australians living in rural and remote locations have a markedly increased risk of developing end-stage kidney disease compared with those living in urban locations. All of the other statements are true.